East West University

Department of CSE

Course Code and Name: **CSE34 (Information System Analysis and Design)**	
Project report: SRS	Group No: 6
Project Name: **Warehouse management System**	
Semester and Year: **Fall 2025**	
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Date of Report Submitted: 27th December 2025	Total Marks Obtained:

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Warehouse management System

Stakeholders Meeting

Objective of the Meeting

The main objective of our stakeholders' meeting is to align all key participants on the goals, scope, and expected outcomes of our WMS project. The key points of the meeting are business requirements, reviewing current workflows, discussing system expectations, and establishing performance indicators that will guide project success.

Stakeholders

- Project Sponsor: Oversees project funding and strategic alignment.
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- Warehouse Manager: Provides insights into daily operations, challenges, and workflow needs.
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- Inventory Controller: Defines stock handling, tracking, and reporting requirements.
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- IT Department: Ensures system integration, data security, and technical support.
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- Procurement Officer: Coordinates supplier and purchase order processes.
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- End Users (Warehouse Staff): Operate the system for receiving, picking, packing, and dispatching.
- Training and Support Team: Develops and delivers user training programs.

Specific Business Needs

1. Reduce manual data entry and increase real-time inventory visibility.
2. Reduce inefficiencies in the warehouse and stock inconsistencies.
3. Improve the performance of on-time delivery and order accuracy.
4. Connect WMS to current accounting and ERP systems.

Desired Outcomes

A. Streamlined warehouse operations with automated workflows.

B. Reduced inventory loss, misplacement, and overstock situations.

C. Improved decision-making through accurate, real-time data reporting.

D. Increased productivity and reduced order processing time.

E. Higher customer satisfaction through faster and error-free deliveries.

Performance Indicators

To measure the success of the WMS implementation, the following KPIs will be tracked:

- Inventory Accuracy Rate: Target ≥ 98%
- Order Fulfillment Time: Reduce by 30% within first 3 months
- Picking Accuracy: Target ≥ 99%
- Warehouse Space Utilization: Improve as much we need
- System Uptime: Maintain 99.5% reliability
- User Adoption Rate: ≥ 90% of staff fully trained and active within 2 weeks of deployment

Current Workflows Overview

Our WMS project aims to automate workflows by introducing barcode/RFID scanning, automated order processing, and real-time stock tracking dashboards and also effective user training planning.

Resources:

https://www.optiproerp.com/what-is-warehouse-management-system/?fbclid=IwY2xjawNoWOZleHRuA2FlbQIxMABicmlkETFLSTVKRVgwN05rd2JBd3Y5AR7ByuKsAp2Eh_9Gv82ATBN0Hr5JOrv_ZcU_q416OTfK3fltV9dCE61CwcetMA_aem_PTX0mZXWg5dvp-TIFD_Rfg

https://www.scribd.com/document/696353656/Warehouse-Management-System?fbclid=IwY2xjawNoWOlleHRuA2FlbQIxMABicmlkETFLSTVKRVgwN05rd2JBd3Y5AR5zJzGsQroe4Ya4Lt_vaHTgIQxjZ1_me4nAshqcoGIJYKv3R_13Ze6l19zmew_aem_hmLwonEHaWJIYXFb3gkEGg

USER STORY

Goal

The primary goal of the Warehouse Management System (WMS) project is to modernize and optimize warehouse operations by introducing automation, real-time visibility, and intelligent decision-making tools. The system aims to reduce manual effort, minimize errors, and enhance overall supply chain efficiency through data-driven management.

Objectives

- **Automate Core Warehouse Functions:** To reduce manual handling and expedite workflows, automate key warehouse operations like receiving, put-away, picking, packing, loading, and dispatch.
- **Boost the accuracy of your inventory:** Utilize barcode or RFID technology to achieve accurate inventory control for real-time updates and error reduction.
- **Improve Space and Location Management:** To increase warehouse accessibility and space utilization, use intelligent location mapping and slotting.
- **Increase the Transparency of Operations:** Give managers access to real-time performance dashboards so they can keep an eye on key performance indicators like stock movement, order accuracy, and fulfillment speed.
- **Shorten Order Fulfillment Time:** To guarantee prompt delivery and satisfied customers, streamline order processing from picking to dispatch.
- **Assure Smooth Data Integration:** To ensure synchronized data flow across sales, logistics, and procurement, integrate WMS with current ERP and accounting systems.
- **Enable Predictive and Sustainable Operations:** Introduce AI-driven analytics to forecast demand, predict stock shortages, and reduce waste. Additionally, promote eco-friendly warehouse operations by minimizing unnecessary movements, optimizing route planning for forklifts, and encouraging paperless documentation.
- **Assisted Picking System:** Introduce a picking system that guides warehouse workers using voice instructions and Augmented Reality (AR) glasses.

The main purposes are:

- Provide full traceability from goods receiving to final dispatch.
- Support optimized material movement and real-time stock visibility.
- Handle invoicing, reconciliation, and reporting automatically.
- Facilitate decision-making with analytics-based insights.
- Foster a culture of continuous improvement and eco-efficiency.

Expected End Users

A wide range of users engaged in supply chain and warehouse operations are intended to benefit from the Warehouse Management System (WMS).

User & System Management

User Authentication & Access Control as an IT/System Administrator want to create and manage user roles and permissions, so that only authorized users can access relevant WMS functions.

Receiving & Quality Control:

Goods Receiving as a Warehouse Staff (Receiver) want to record received goods using barcode/RFID scanning, so that stock is accurately added to inventory.

Quality Inspection as a Quality Control Officer want to inspect received goods and mark them as approved, rejected, or quarantined, so that only compliant items enter inventory.

Inventory & Location Management

Real-Time Inventory Tracking as an Inventory Officer want to view real-time stock levels and locations, So that inventory accuracy is maintained.

Storage Location Assignment as a Warehouse Staff want to receive system-suggested storage locations, so that warehouse space is optimized.

Order Processing & Picking

Order Creation as a Supply Chain Officer want to create outbound orders in the WMS, so that picking and shipping can be initiated.

Assisted Picking System as a Picker want to receive voice or AR-based picking instructions,
So that picking speed and accuracy improve.

Packing & Shipping

Packing Verification as a Packing Staff want to verify picked items before packing, so that order accuracy is ensured.

Shipment Dispatch as a Dispatch Officer want to dispatch orders and update shipment status, so that customers can track deliveries.

Returns Management

Returns Processing as a Warehouse Staff want to receive returned goods and record return reasons, So that inventory remains accurate.

EPIC 7: Analytics & Reporting

Operational Dashboard as a Warehouse Manager want to view real-time dashboards,
So that monitoring warehouse performance easy.

Inventory Analytics as a Supply Chain Manager want to receive demand forecasts and shortage alerts, so that stockouts are reduced.

Customers:

In advanced WMS setups with customers may track order status, inventory levels, and shipment schedules through a dedicated interface,

Project features

1. Receiving Management

The receiving module handles all inbound goods entering the warehouse. It ensures accurate recording, inspection, and storage of materials.

Key Features:

- Record goods received against purchase orders
- Barcode/RFID-based item scanning
- Quantity and batch verification
- Quality inspection and approval/rejection handling
- Automatic generation of receiving reports
- Assignment of appropriate storage locations

Benefits:
Improves accuracy of inbound stock and ensures full traceability from receipt to storage

2. Inventory Management

This module maintains real-time visibility of inventory levels across the warehouse.

Key Features:

- Real-time stock tracking
- Support for FIFO and LIFO stock rotation
- Batch, lot, and expiry date management
- Automatic stock updates on every transaction
- Stock adjustment and cycle count support

Benefits:
Reduces inventory discrepancies and prevents stockouts or overstocking.

3. Storage & Location Management

Optimizes warehouse space and improves item accessibility.

Key Features:

- Intelligent location mapping
- System-recommended put-away locations
- Capacity-based slotting
- Fast item location search

Benefits:
Improves space utilization and reduces material handling time.

4. Order Management

Manages outbound customer and internal orders efficiently.

Key Features:

- Order creation and modification

- Order prioritization (urgent, standard)
- Real-time order status tracking
- Integration with sales and procurement systems

Benefits:
Ensures smooth and timely order processing.

5. Assisted Order Picking

Enhances picking speed and accuracy using modern technologies.

Key Features:

- Automatic pick list generation
- Barcode confirmation during picking
- Voice-guided picking instructions
- Augmented Reality (AR) assisted navigation

Benefits:
Reduces picking errors and increases workforce productivity

6. Packing & Shipping

Ensures safe packing and accurate dispatch of goods.

Key Features:

- Pick verification before packing
- Packing confirmation and labeling
- Shipping label and document generation
- Shipment status updates and tracking

Benefits:
Improves order accuracy and customer satisfaction.

7. Returns Management

Handles customer and internal returns efficiently.

Key Features:

- Return receipt and inspection
- Return reason recording
- Restocking or disposal processing
- Inventory updates after returns

Benefits:
Maintains accurate stock records and supports reverse logistics.

8. Analytics & Reporting

Provides insights into warehouse performance and trends.

Key Features:

- Real-time operational dashboards
- Daily, weekly, and monthly reports
- Key performance indicators (KPIs) tracking
- Demand forecasting and trend analysis

Benefits:
Supports data-driven decision-making and continuous improvement.

9. System Integration

Ensures seamless data exchange with other enterprise systems.

Key Features:

- Integration with ERP and accounting systems
- Automated invoicing and reconciliation
- API-based data synchronization

Benefits:
Eliminates data duplication and improves financial accuracy.

Assumptions and Constraints

Assumptions

The following assumptions are made to ensure successful development and implementation of the Warehouse Management System (WMS):

1. **User Training Availability**
   All warehouse staff, managers, and relevant departments will receive adequate training on the WMS, including barcode/RFID scanning, dashboards, assisted picking tools, and reporting features.

2. **Infrastructure Readiness**
   The warehouse has sufficient hardware and network infrastructure, including stable internet connectivity, computers, barcode scanners, and mobile devices required to operate the WMS effectively.
3. **Accurate Initial Data Migration**
   Existing inventory, item master data, supplier data, and location data are assumed to be accurate, verified, and cleaned before migration into the new system.
4. **Management and Staff Support**
   Senior management and operational staff are fully supportive of the WMS implementation and will cooperate during system adoption and process changes.
5. **System Integration Feasibility**
   The WMS can technically integrate with existing ERP, accounting, and procurement systems using APIs, middleware, or file-based data exchange.
6. **Operational Stability During Implementation**
   No major changes in warehouse layout, business rules, or operational processes will occur during the development and deployment phase.
7. **Reliable Power and Network Supply**
   Continuous electrical power and network connectivity will be available to ensure uninterrupted system operation.
8. **Timely Technical Support**
   The development and support team will provide timely bug fixes, maintenance, and updates within agreed service-level timelines.

Constraints

The following constraints may limit or influence the scope, timeline, and functionality of the WMS project:

1. **Budget Constraints**
   Financial limitations may restrict the implementation of advanced features such as full RFID automation, AI-based robotics, or large-scale AR deployment.
2. **Project Timeline**
   The system must be designed, developed, tested, and deployed within a fixed timeframe.
3. **Dependency on External Teams**
   Delays from third-party vendors, ERP teams, or hardware suppliers may impact project schedules.
4. **User Adaptation Challenges**
   Some users may require additional training and support due to limited technical experience, potentially affecting early adoption.
5. **Data Migration Risks**
   Errors or inconsistencies may occur during data migration from legacy systems, leading to temporary inaccuracies in inventory records.

6. **Warehouse Layout Limitations**
   Physical constraints of the warehouse may limit optimal placement of scanners, workstations, sensors, or network devices.
7. **Regulatory and Policy Compliance**
   The system must comply with organizational policies, data protection laws, and industry regulations related to inventory, finance, and customer data.
8. **Operational Downtime During Deployment**
   Temporary interruptions or reduced efficiency may occur during system installation, testing, or go-live phases.

Feasibility Study for WMS (Raw Material Warehouse)

1. Technical Feasibility

The Warehouse Management System is technically feasible because it uses available software, hardware, and network technology to manage daily warehouse activities like receiving, inventory, order picking, shipping, and reporting. The IT or software engineer can configure the WMS and connect it with barcode or RFID scanners for tracking materials in real time. The existing computers, scanners, and internet connection can easily support the system. Cloud-based databases can safely store and share information between the warehouse and the main ERP system. Since the system analyst and software engineer already have the required skills, developing and maintaining the WMS is practical.

2. Economic Feasibility

The Warehouse Management System is economically feasible because its cost is reasonable compared to the benefits it provides. The main expenses include hardware like barcode scanners, handheld devices, software licenses, and staff training. In return, the system helps reduce labor hours, minimize picking and shipping errors, and improve inventory accuracy. For example, if the WMS reduces material handling time by 10% and saves around $15,000 every year, the total investment can be recovered within two to three years. These long-term savings and increased efficiency show that implementing the WMS is a cost-effective and financially smart decision for the organization.

3. Operational Feasibility

The Warehouse Management System is operationally feasible because it fits well with the warehouse's daily activities and staff routines. It replaces manual work with a faster and more accurate digital process. For example, goods can be entered digitally instead of using paper logs, and inventory updates can be done in real time through tablets. The system also generates automatic pick lists to reduce mistakes and confusion. Managers can easily track stock movement and warehouse performance through reports. With short training, the warehouse supervisor and staff can quickly adapt to the new system, improving accuracy, speed, and overall control of operations.

4. Legal Feasibility

The Warehouse Management System is legally feasible because it follows all rules and standards for handling business and supplier data. The system stores vendor and shipment information securely, ensuring that no unauthorized person can access it. Since the WMS only manages operational data, such as stock levels and materials, there is no risk of violating personal privacy laws. All software and hardware used in the system are properly licensed and follow the company's IT policies. As a result, there are no expected legal or compliance issues, making the WMS safe and lawful to use within the organization's operations.

PROJECT CHARTER

Project Title	Warehouse management System	**Project Manager**	Sadia Zaman

Project Start Date	26/10/2025	**Project End Date**	18/12/2025	**Project Sponsor**	Ahmed Adnan

Business Need

The main objective of our stakeholders' meeting is to align all key participants on the goals, scope, and expected outcomes of our WMS project. The points of the meeting are business requirements, reviewing current workflows, discussing system expectations, and establishing performance indicators that will guide project success.

Project Scope

- Streamlined warehouse operations with automated workflows.
- Reduced inventory loss, misplacement, and overstock situations.
- Improved decision-making through accurate, real-time data reporting.
- Increased productivity and reduced order processing time.
- Higher customer satisfaction through faster and error-free deliveries.

Deliverables

- Receiving
- Inventory
- Order Picking
- Packing & Shipping
- Returns Handling
- Analytics and Reporting

Risks and Issues

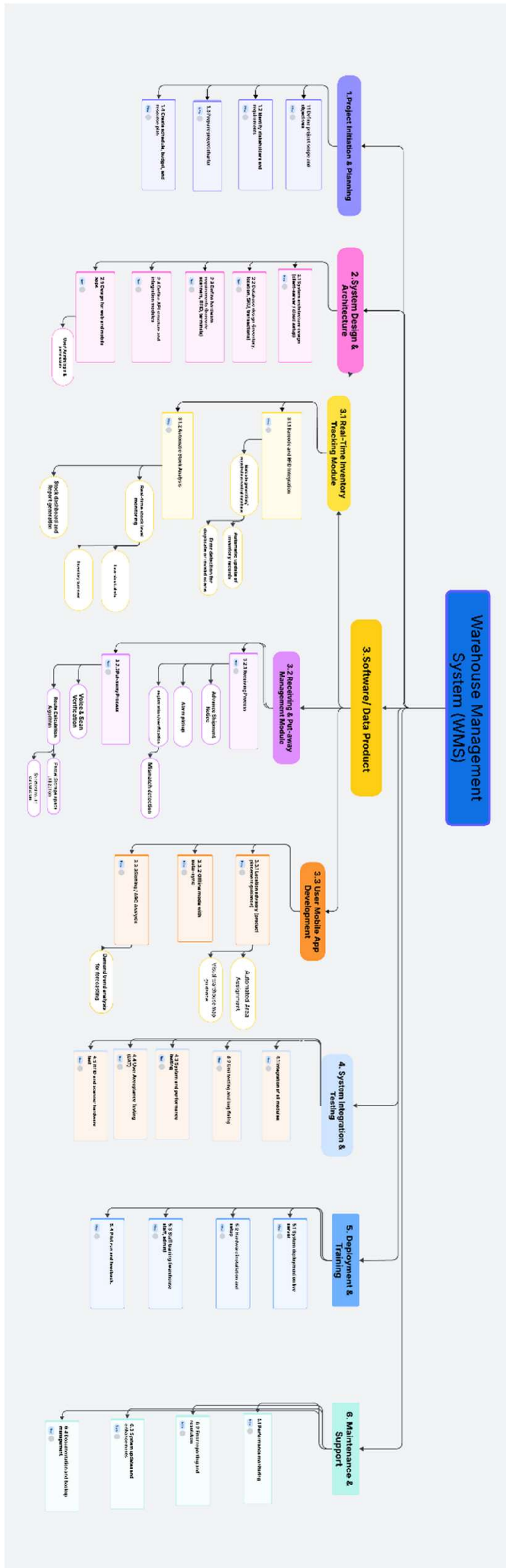
- Cost constraints may limit advanced features like full RFID automation.
- The system must be deployed within a set timeframe like 3–4 months.
- Delays from external teams can slow down the project schedule.
- Extra training and support may be needed for staff to use the system properly
- Errors may occur when moving old data into the new system, which could cause inaccuracies.
- The system must follow company policies and data protection laws.
- Temporary interruptions may happen while installing or testing the system.

Assumptions/Dependencies

1. All warehouse staff will receive training on the new WMS. It is assumed users will be trained to use scanners, dashboards, and reporting tools.
2. The existing hardware and network infrastructure are sufficient. The warehouse already has stable internet, barcode scanners, and computers to run the system.
3. Data on raw materials and stock levels is accurate at the start of system migration. The initial inventory data used to populate the WMS is clean and validated.
4. Power and network connectivity will be reliable. The WMS depends on stable network and electrical supply to function continuously
5. The development team will provide timely technical support. Bugs or issues will be fixed within agreed service levels.

Financials		
Budget to complete this project is $3000.		
Milestones Schedule		
Milestone	Target Completion Date	Actual Date
Upload all Global IT Projects to the tool	23/10/2025	26/10/2025
Complete UAT testing for the tool	18/12/2025	18/12/2025
Project Team		
Project Manager	Sadia Zaman	Overall project oversight and coordination
System Analyst	Nabila Binti Hossain	Requirement gathering and Process documents
Software Engineer	Abu Sufian	WMS configuration, integration and testing
Software Engineer	Kowser Ahamed	WMS configuration, integration and testing
Supervisor	Mehjabin Alam jeba	Process validation and end-user feedback
Quality Assurance	M Sadman Hossain Bhiuyan	System testing and performance verification
Approval/Review Committee		
Sponsor	Ahmed Adnan	
Business Division Head	Mujahedul Islam	
Business Unit Head	Sajid Ikbal	

WBS Diagram

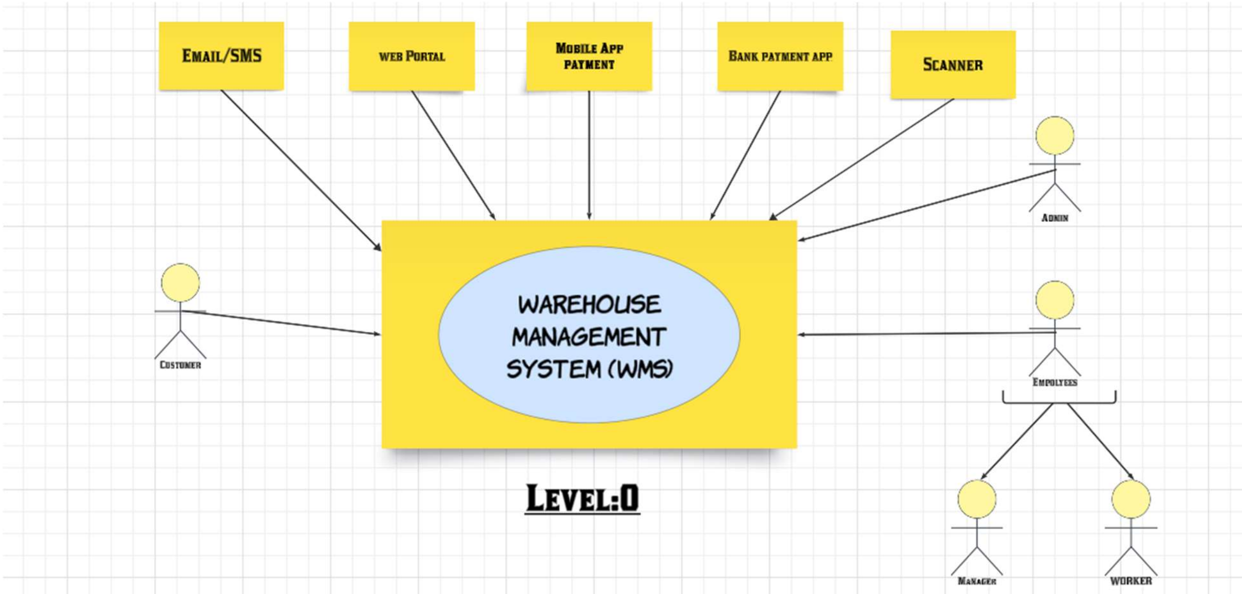

Warehouse Management System (WMS)
1.Project Initiation & Planning
2.System Design & Architecture
3.Software/ Data Product
3.1 Real-Time Inventory Tracking Module
3.2 Receiving & Put-away Management Module
3.3 User Mobile App Development
4. System Integration & Testing
5. Deployment & Training
6. Maintenance & Support

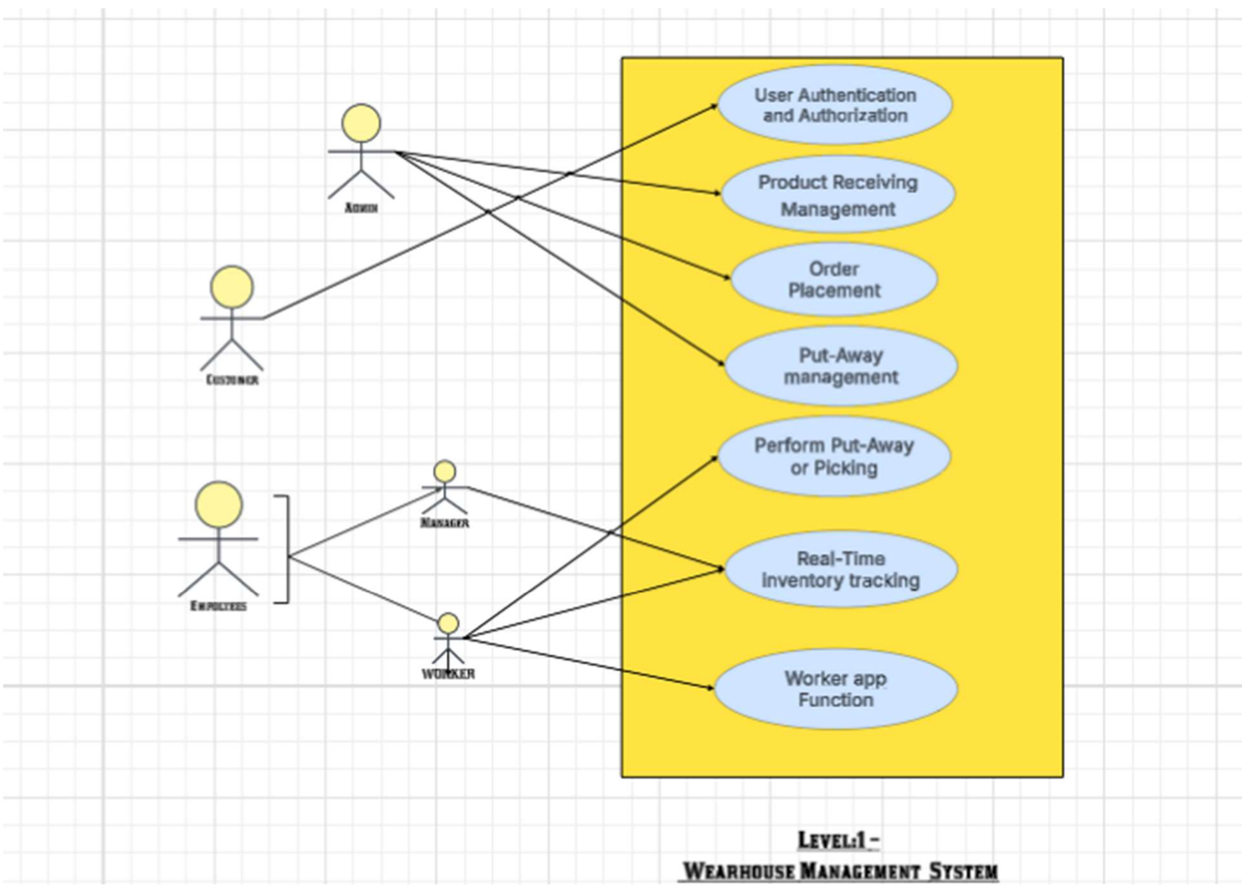
Cost Estimation

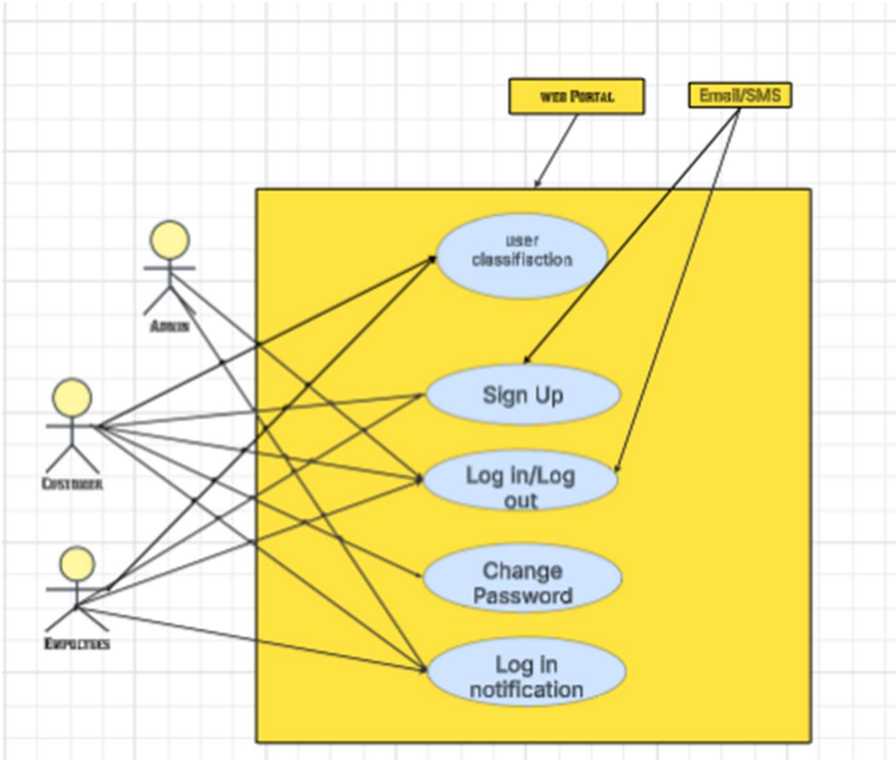
component	Specific Feature	optimistic	most likely	pessimistic	Estimated (E) = (O+M+P)/3	pert
1. Requirement Analysis	Requirement gathering & documentation	25,000	30,000	40,000	31,667	30833.3333
	Stakeholder meetings & approvals	15,000	20,000	30,000	21,667	20833.3333
2. System Design & Architecture	System architecture design (clientserver/cloud)	40,000	55,000	75,000	56,667	55833.3333
	Database design (inventory, SKU, transactions)	50,000	65,000	90,000	68,333	66666.6667
	Hardware requirement & integration (barcode, RFID)	55,000	70,000	100,000	75,000	66666.6667
	API & module design	35,000	50,000	70,000	51,667	72500
	UI/UX design (web & mobile apps)	40,000	60,000	85,000	61,667	50833.3333
3. Real-Time Inventory Tracking Module	Barcode & RFID Integration	60,000	80,000	1,10,000	83,333	60833.3333
	Slotting / ABC Analysis	45,000	65,000	90,000	66,667	65833.3333
	Stock Analysis Dashboard	40,000	55,000	80,000	58,333	56666.6667
4. Receiving & Put-away Management Module	Receiving process (registration & barcode/RFID check)	45,000	60,000	85,000	63,333	61666.6667
	Put-away process (task interleaving, route, voice scan, app guidance)	55,000	75,000	1,00,000	76,667	750000

5. Worker Mobile App Development	App UI/UX for operators	30,000	45,000	65,000	46,667	45833.3333
	Login & user role management	25,000	35,000	50,000	36,667	35833.3333
	Real-time sync with central WMS	40,000	55,000	80,000	58,333	56666.6667
	Location advisory (placement guidance)	25,000	40,000	60,000	41,667	40833.3333
	Voice command & scanning interface	30,000	45,000	70,000	48,333	46666.6667
	Notification & alert system	15,000	25,000	35,000	25,000	56666.6667
6. System Integration & Testing	Integration of all modules	40,000	60,000	80,000	60,000	40833.3333
	Unit testing & bug fixing	25,000	40,000	60,000	41,667	46666.6667
	System & performance testing	30,000	45,000	70,000	48,333	25000
	User Acceptance Testing (UAT)	15,000	25,000	40,000	26,667	60000
	RFID & scanner hardware testing	20,000	35,000	50,000	35,000	40833.3333
. Deployment & Training	System deployment on live server	20,000	30,000	45,000	31,667	46666.6667
	Hardware installation & setup	15,000	25,000	35,000	25,000	25833.3333
	Staff training & pilot run	25,000	40,000	60,000	41,667	35000
:. Maintenance & Support	Performance monitoring	10,000	15,000	25,000	16,667	30833.3333
	Error reporting & resolution	10,000	15,000	20,000	15,000	25000
	System updates & enhancements	15,000	25,000	35,000	25,000	40833.3333
	Documentation & backup management	10,000	15,000	25,000	16,667	15833.3333

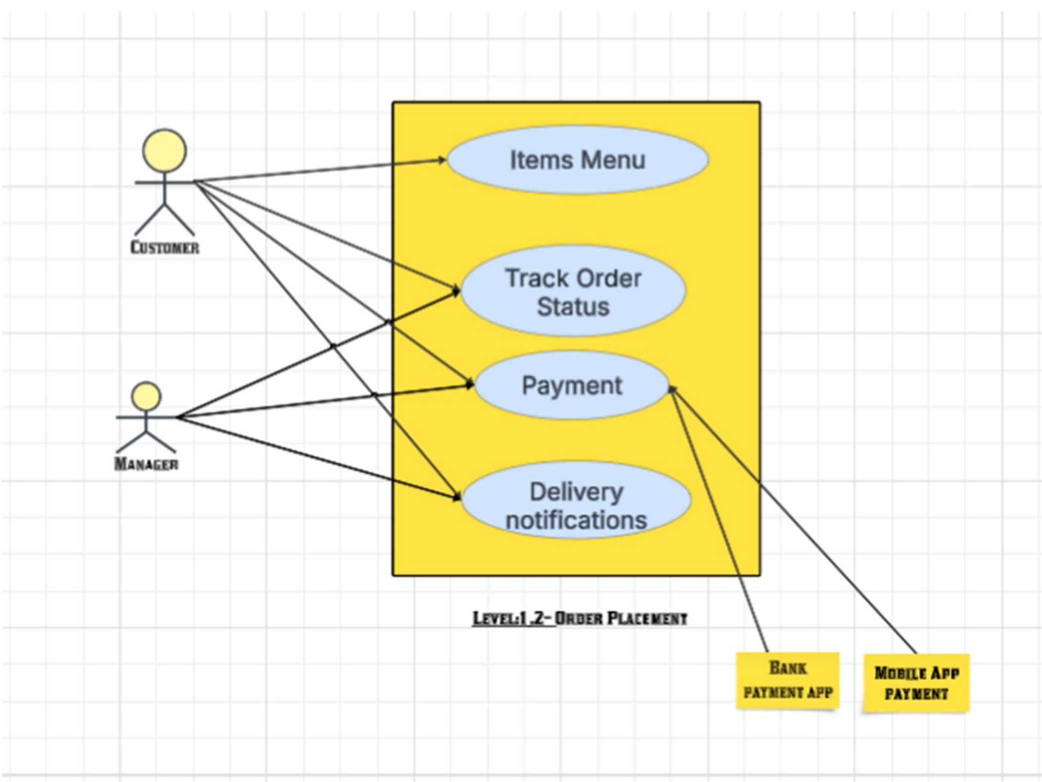
USE CASE


EMAIL/SMS
WEB PORTAL
MOBILE APP PAYMENT
BANK PAYMENT APP
SCANNER
ADMIN
WAREHOUSE MANAGEMENT SYSTEM (WMS)
CUSTOMER
EMPOLYEES
MANAGER
WORKER
LEVEL:0

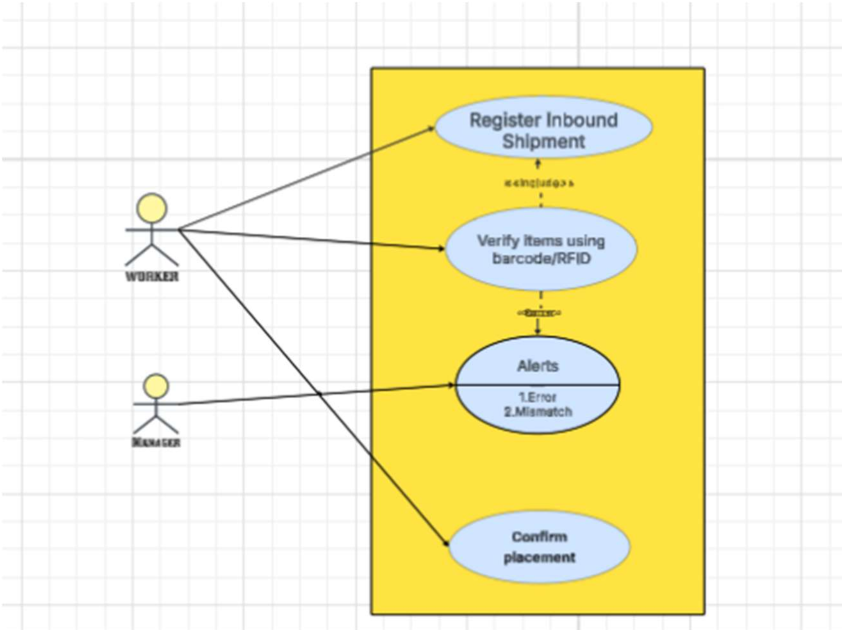

User Authentication and Authorization
Product Receiving Management
Order Placement
Put-Away management
Perform Put-Away or Picking
Real-Time Inventory tracking
Worker app Function
ADMIN
CUSTOMER
EMPOLYEES
MANAGER
WORKER
LEVEL:1 -
WEARHOUSE MANAGEMENT SYSTEM



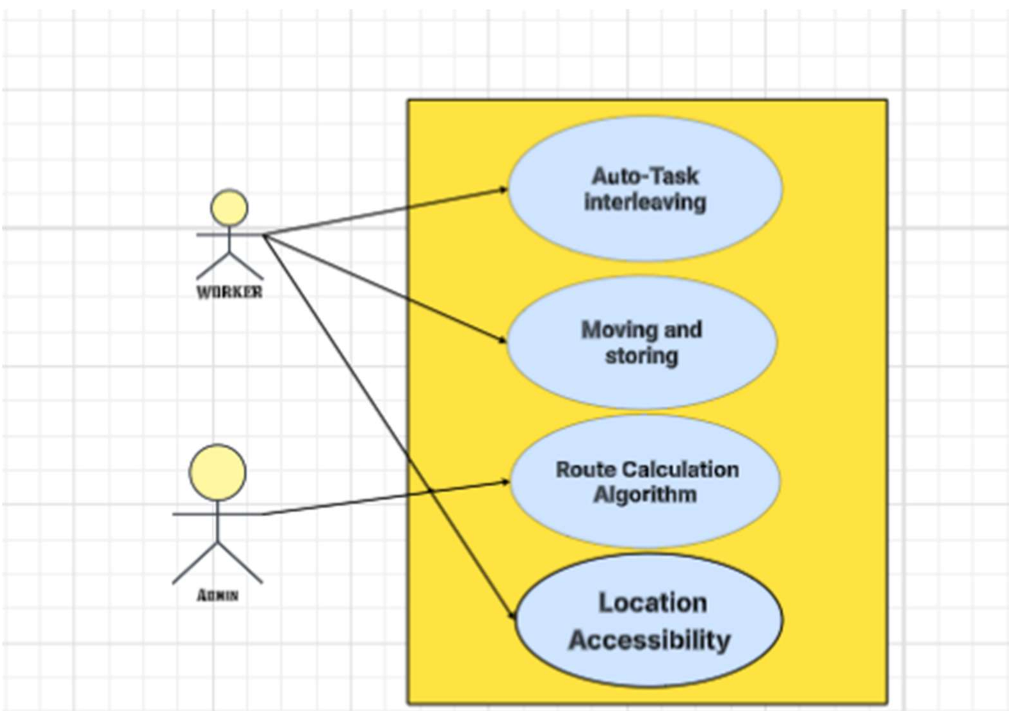
Level:1 .1 -
User Authentication and Authorization



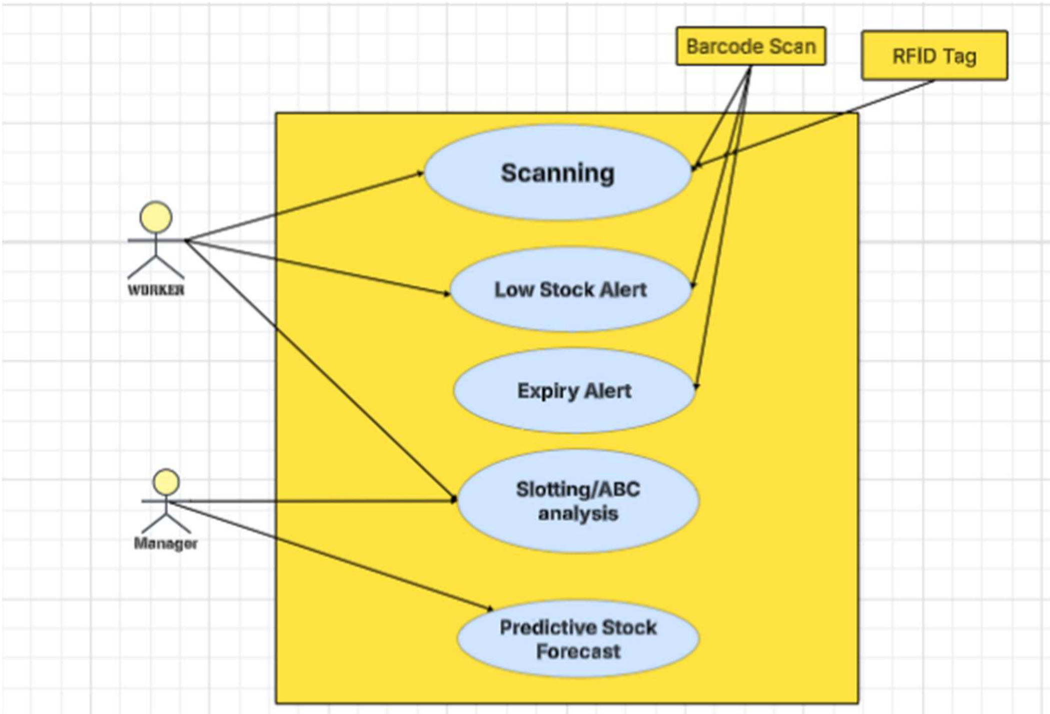
Level:1 .2- Order Placement



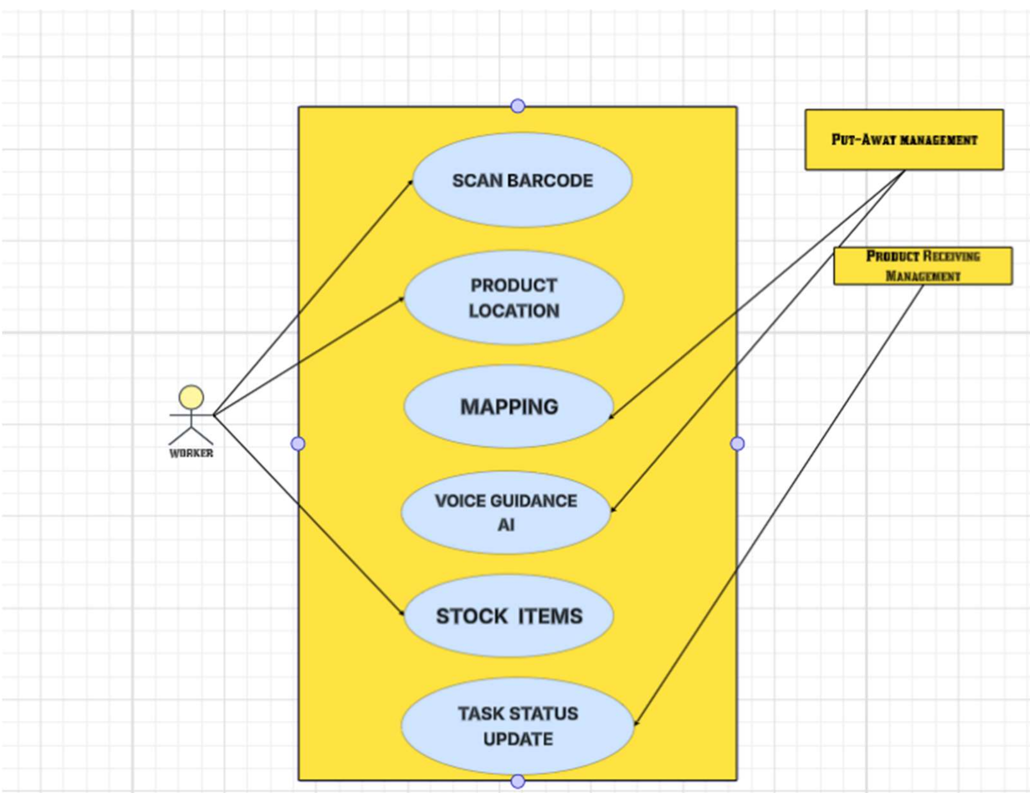
LEVEL:1 .3- PRODUCT RECEIVING MANAGEMENT



LEVEL:1 .4- PUT-AWAY MANAGEMENT

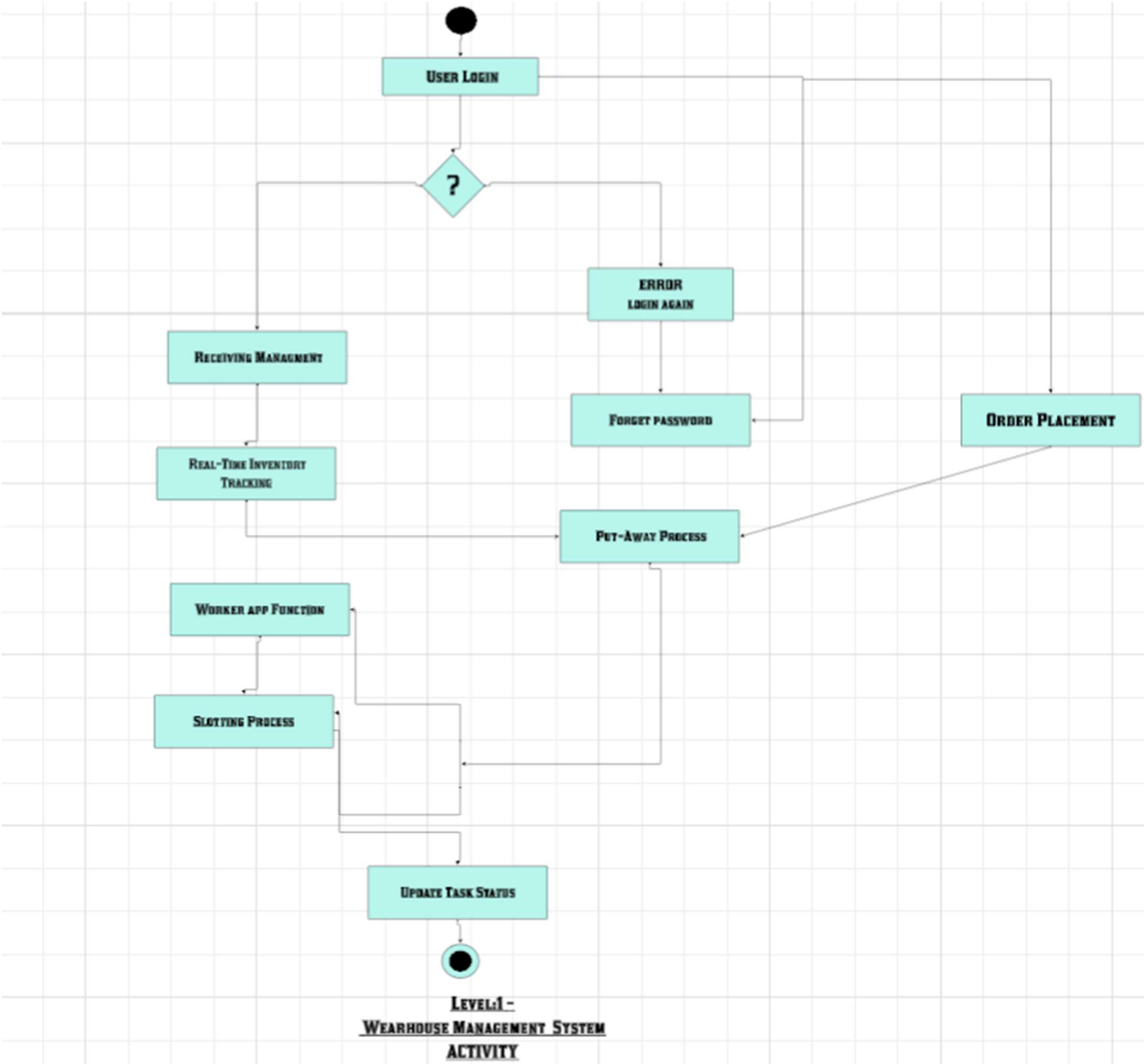


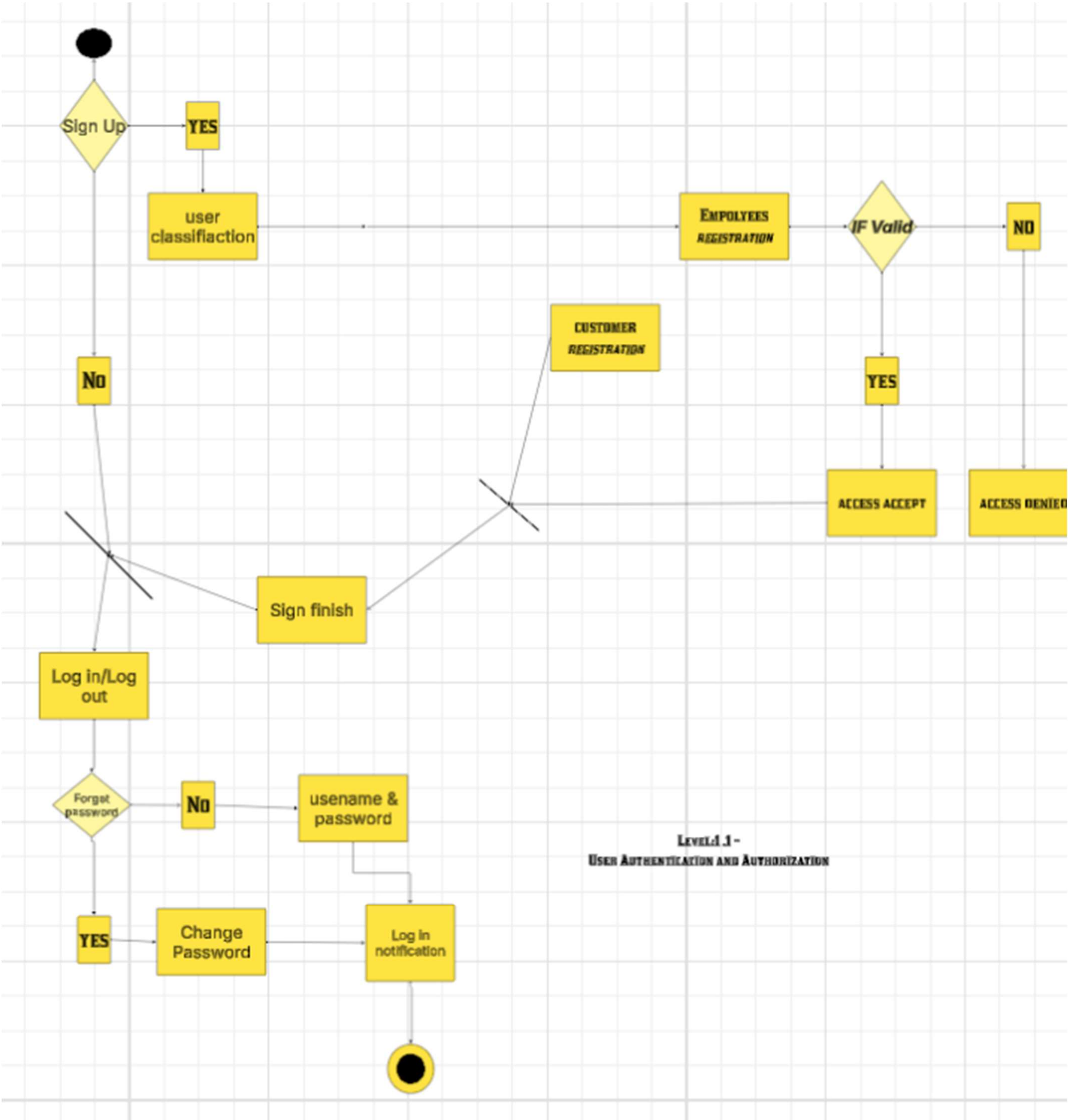
1.5- REAL -TIME INVENTORY TRACKING

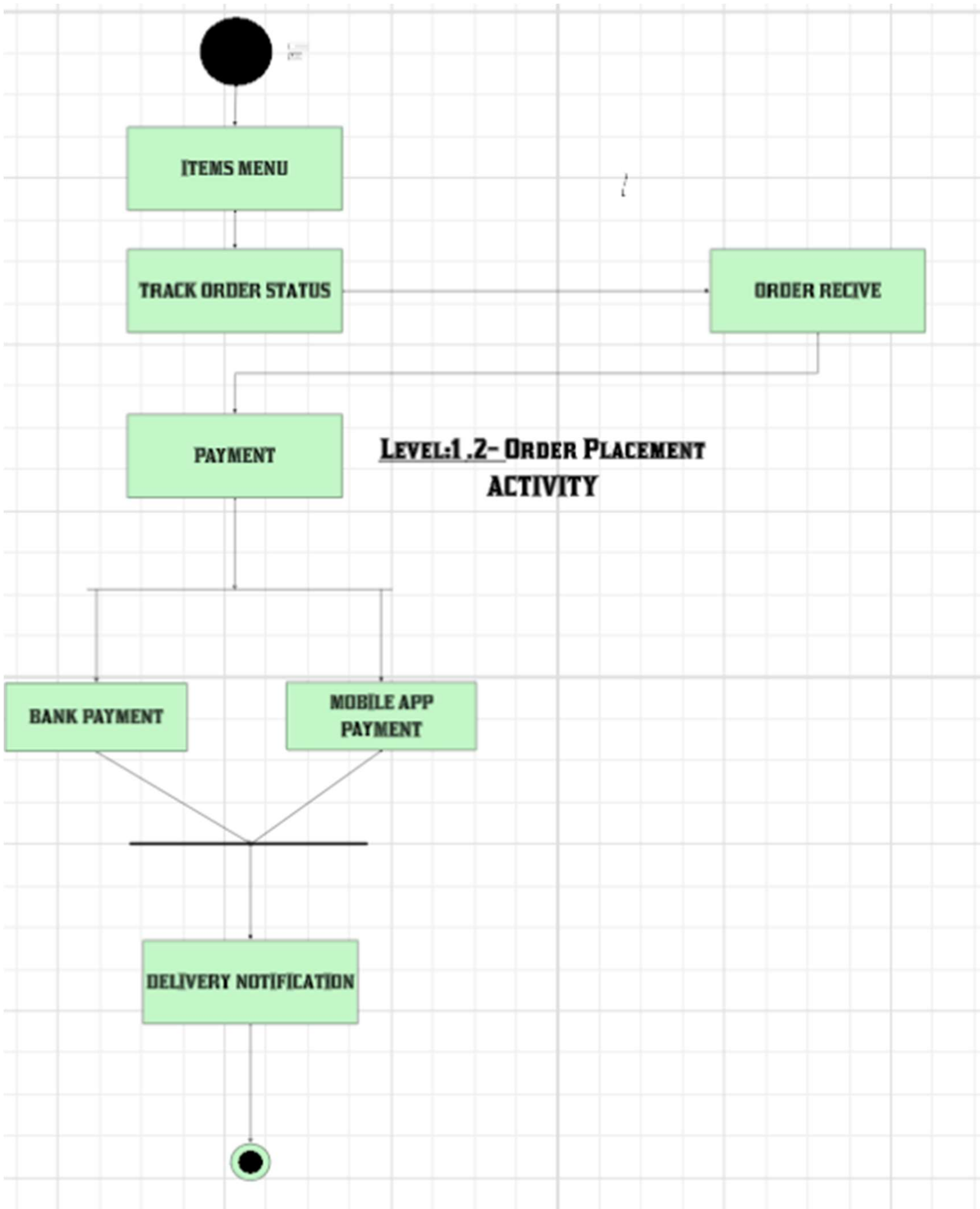


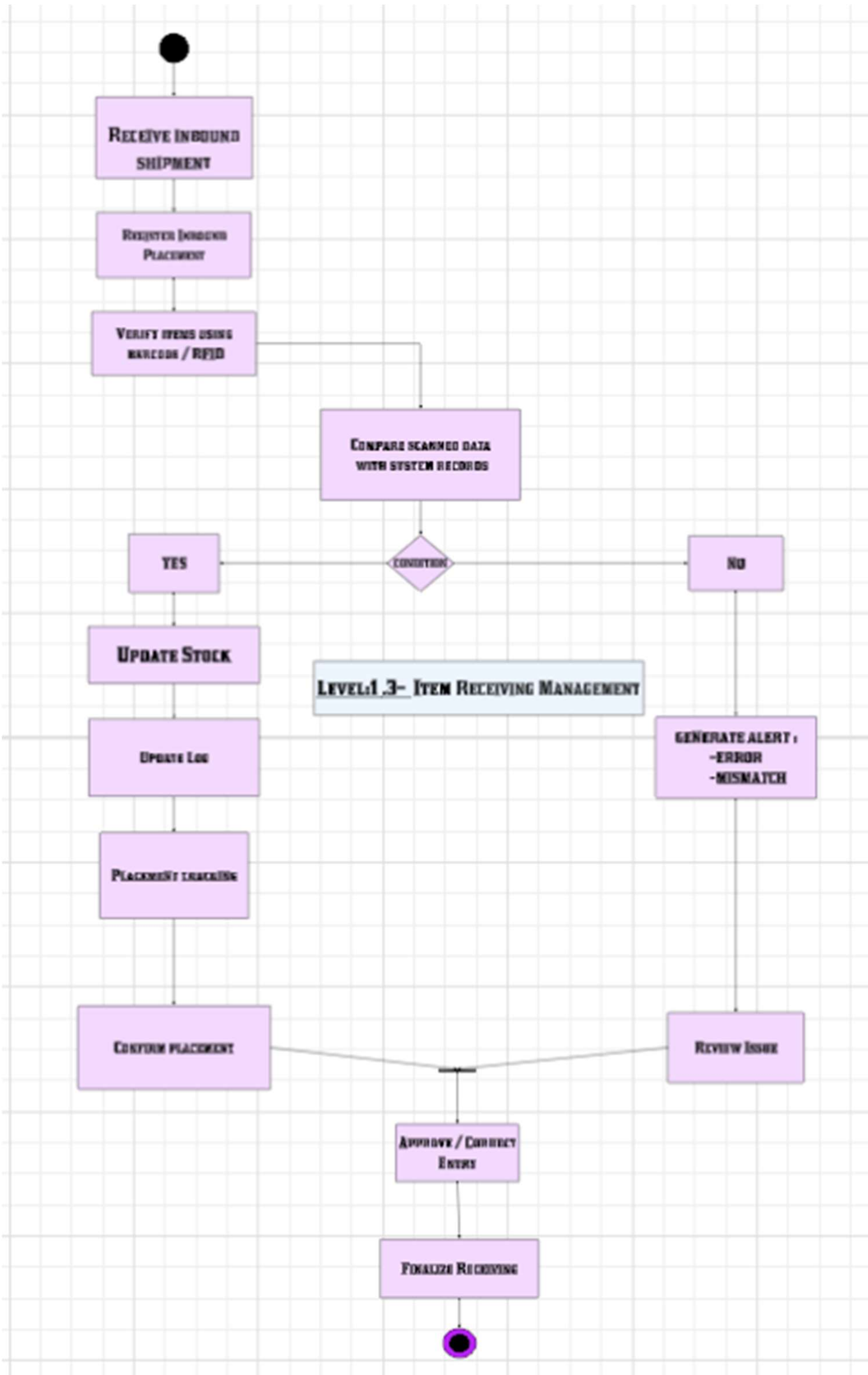
1.6- WORKER MOBILE APP

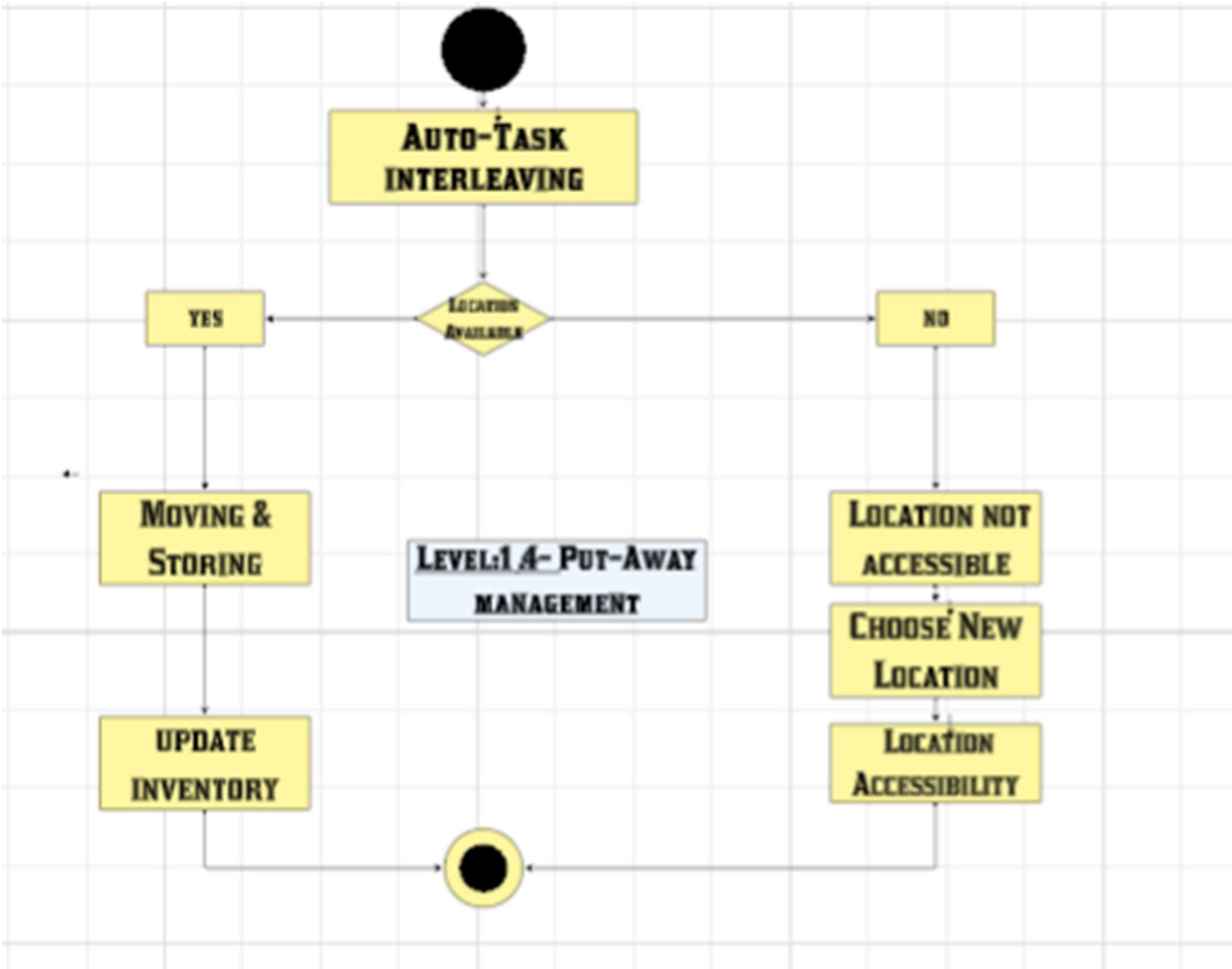
Activity diargram

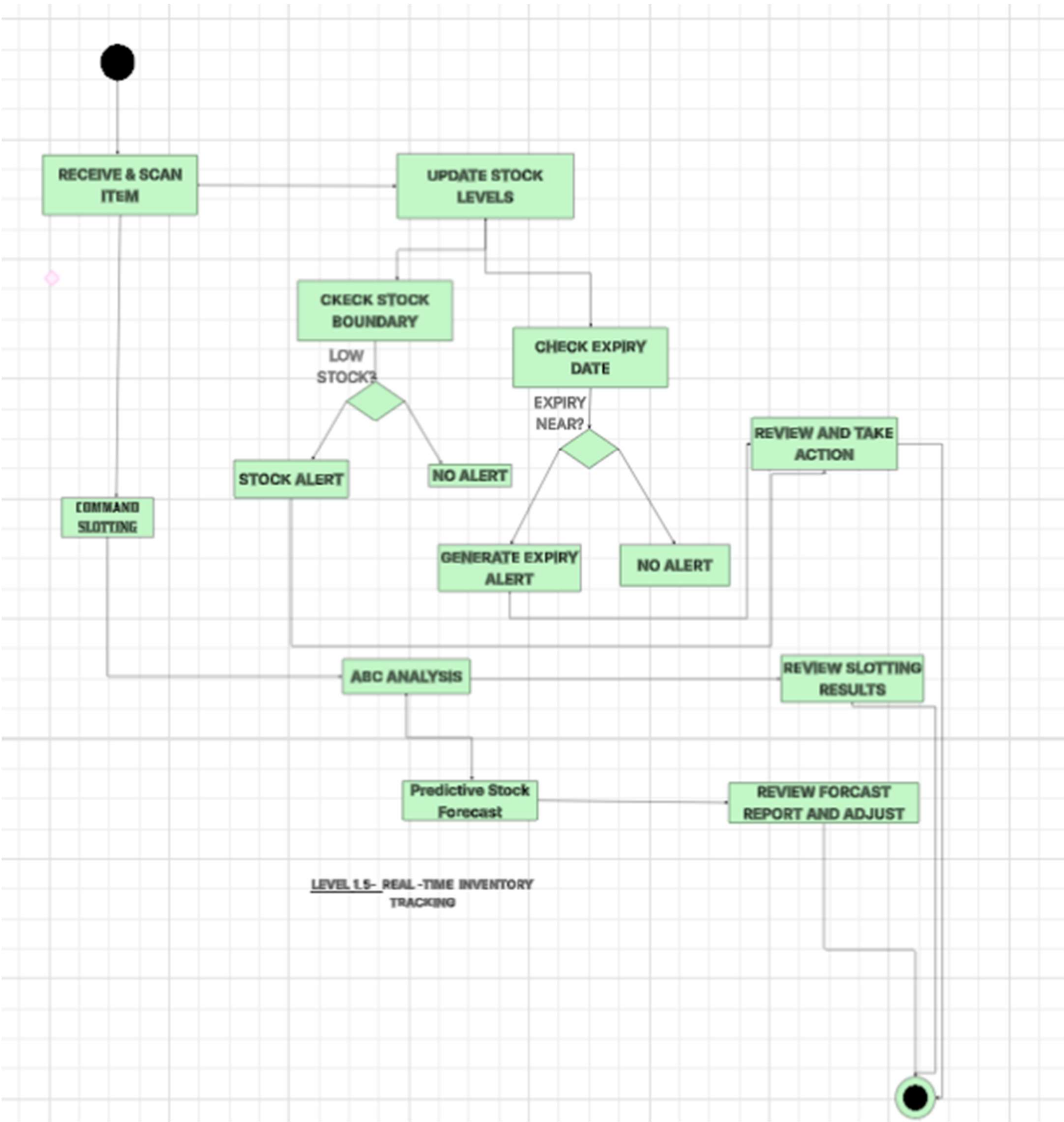



Sign Up
YES
user classiflaction
EMPOLYEES REGISTRATION
IF Valid
NO
CUSTOMER REGISTRATION
No
YES
ACCESS ACCEPT
ACCESS DENIED
Sign finish
Log in/Log out
Forget password
No
usename & password
LEVEL:1.1 -
USER AUTHENTICATION AND AUTHORIZATION
YES
Change Password
Log in notification

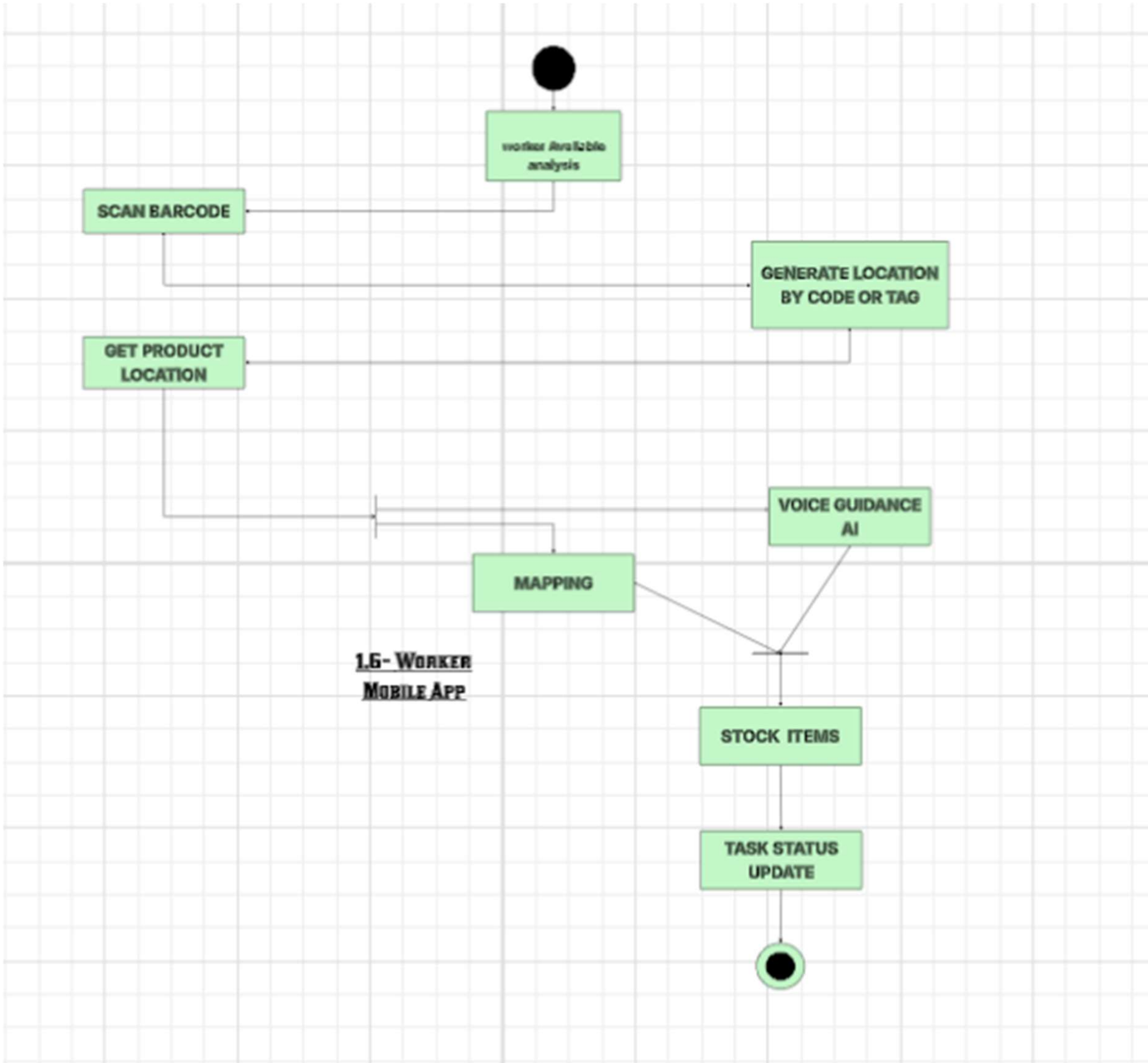

ITEMS MENU
TRACK ORDER STATUS
ORDER RECIVE
PAYMENT
LEVEL:1 .2- ORDER PLACEMENT ACTIVITY
BANK PAYMENT
MOBILE APP PAYMENT
DELIVERY NOTIFICATION


Receive Inbound Shipment
Register Inbound Placement
Verify items using barcode / RFID
Compare scanned data with system records
YES
NO
Update Stock
Level:1.3- Item Receiving Management
Update Log
Generate Alert :
-Error
-Mismatch
Placement tracking
Confirm placement
Review Issue
Approve / Correct Entry
Finalize Receiving


Auto-Task interleaving
Location Available
YES
NO
Moving & Storing
Level:1 A- Put-Away management
Location not accessible
Choose New Location
Update inventory
Location Accessibility

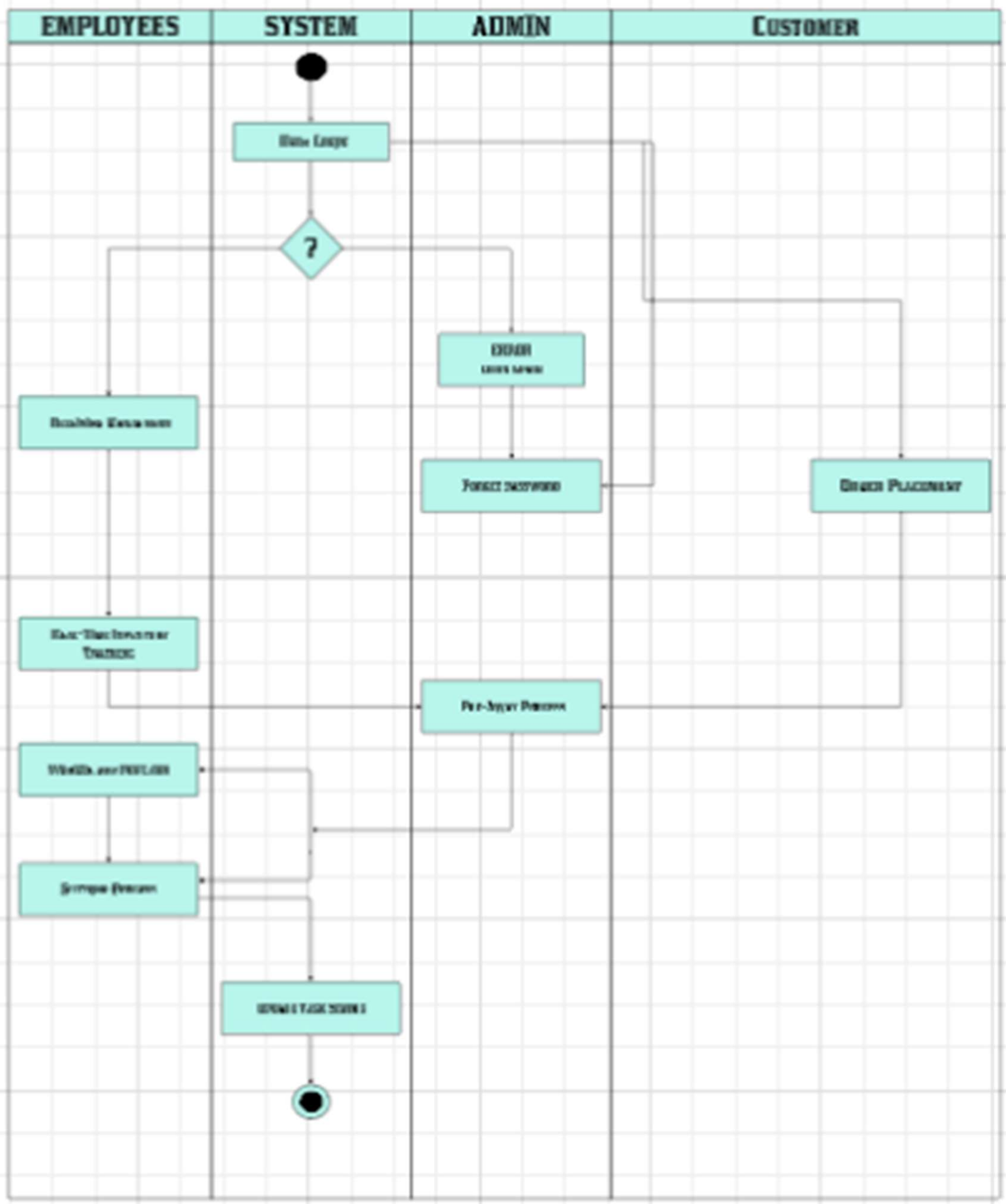


LEVEL 1.5- REAL -TIME INVENTORY TRACKING

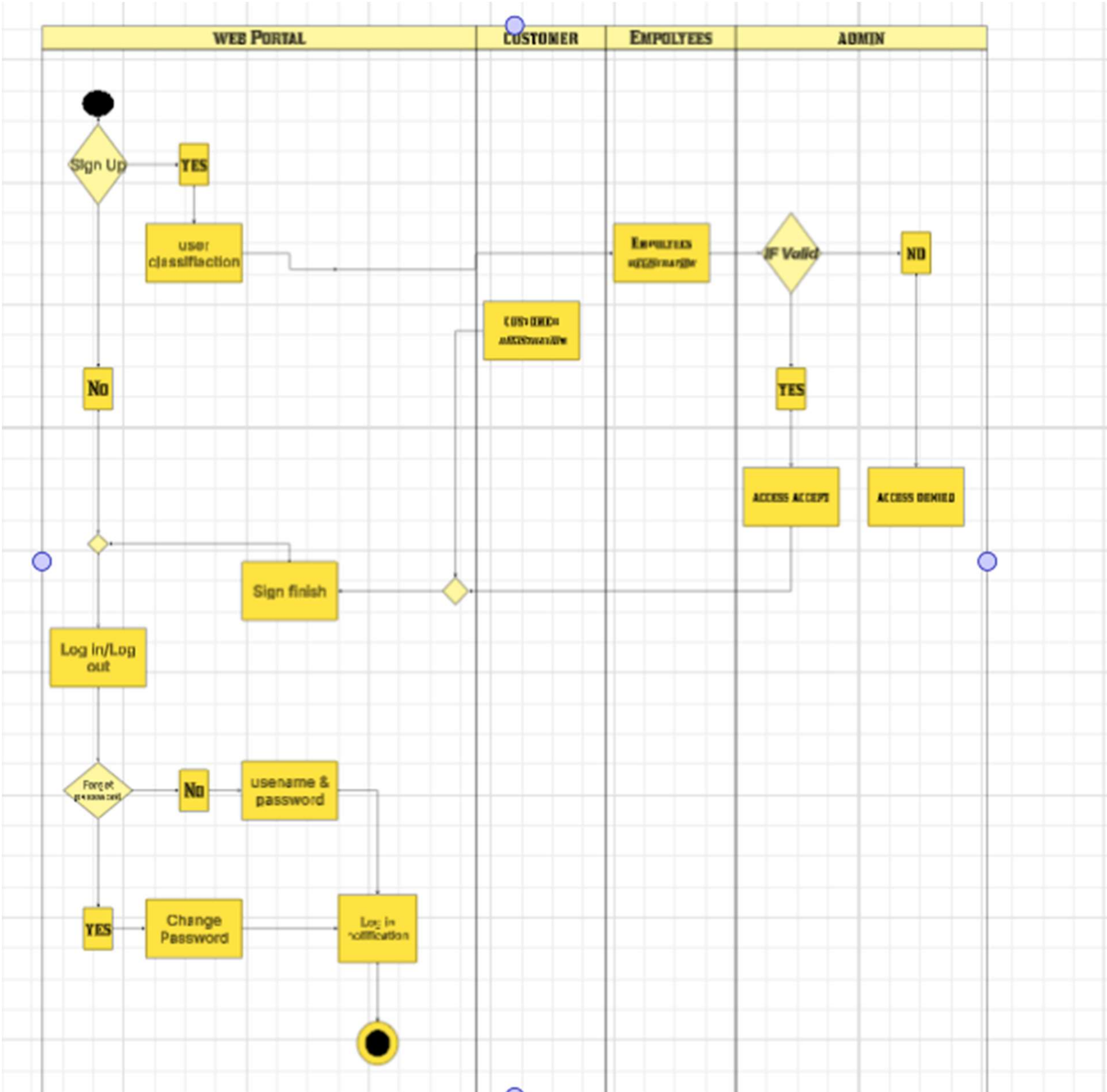


1.6- Worker Mobile App

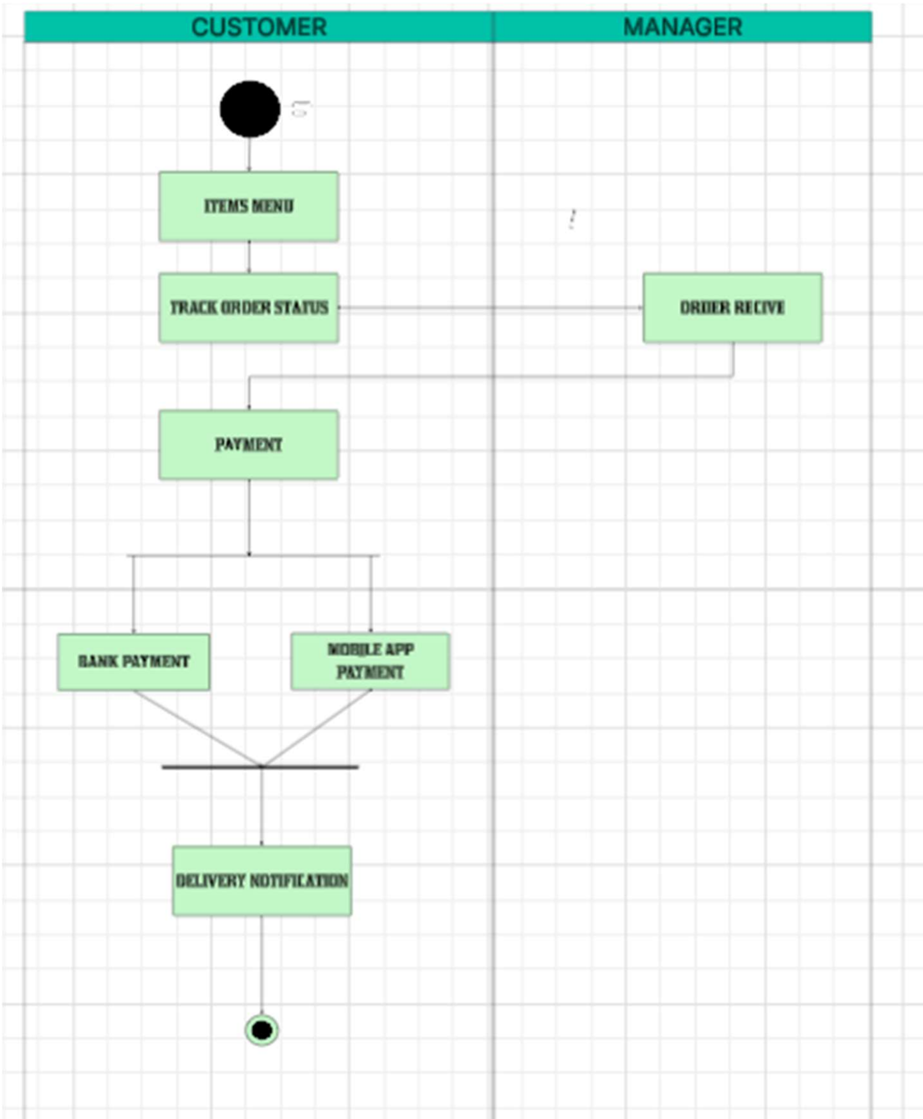
Swimline diagram



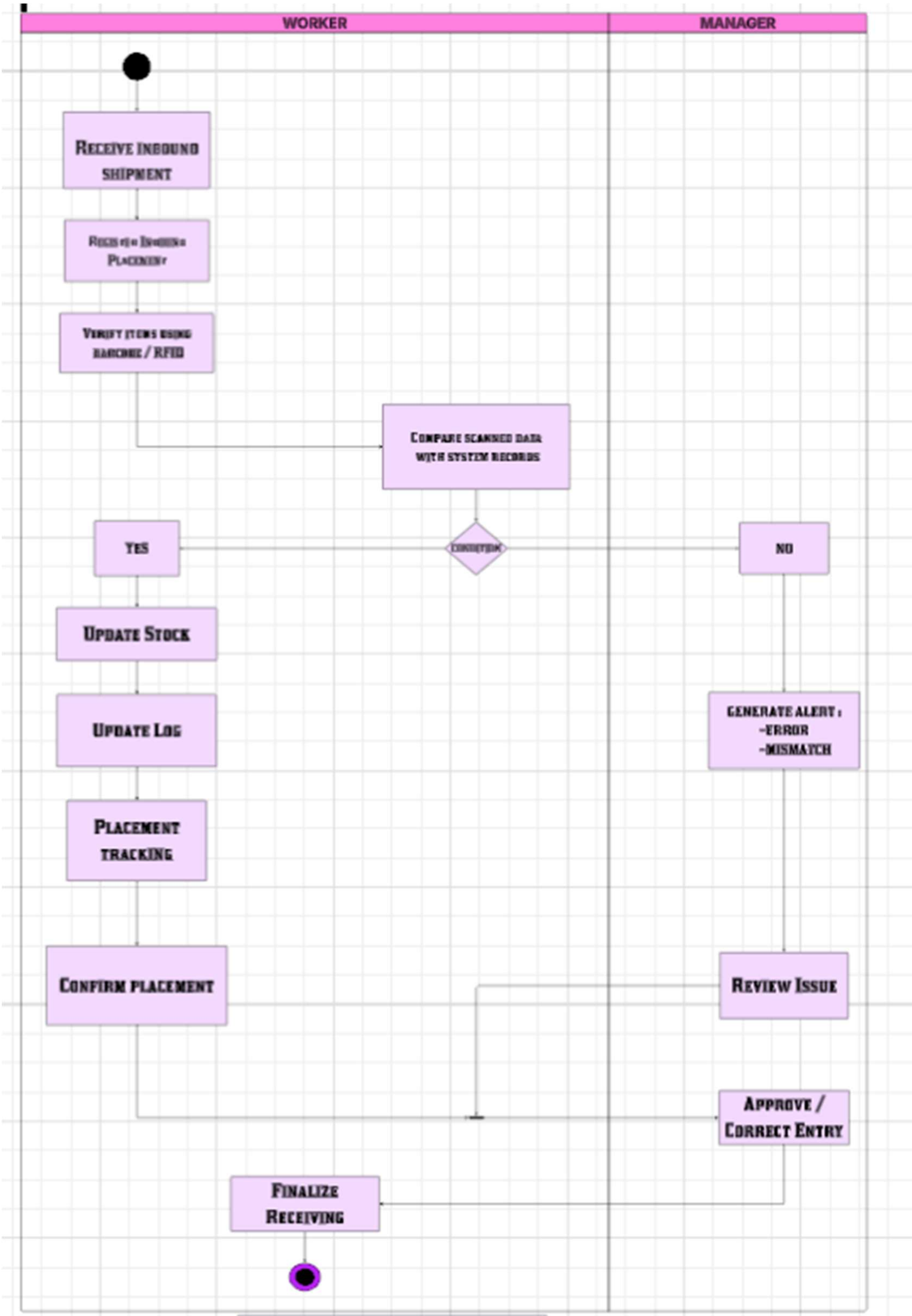
LEVEL:1 -
WEARHOUSE
MANAGEMENT SYSTEM
SWIMLINE



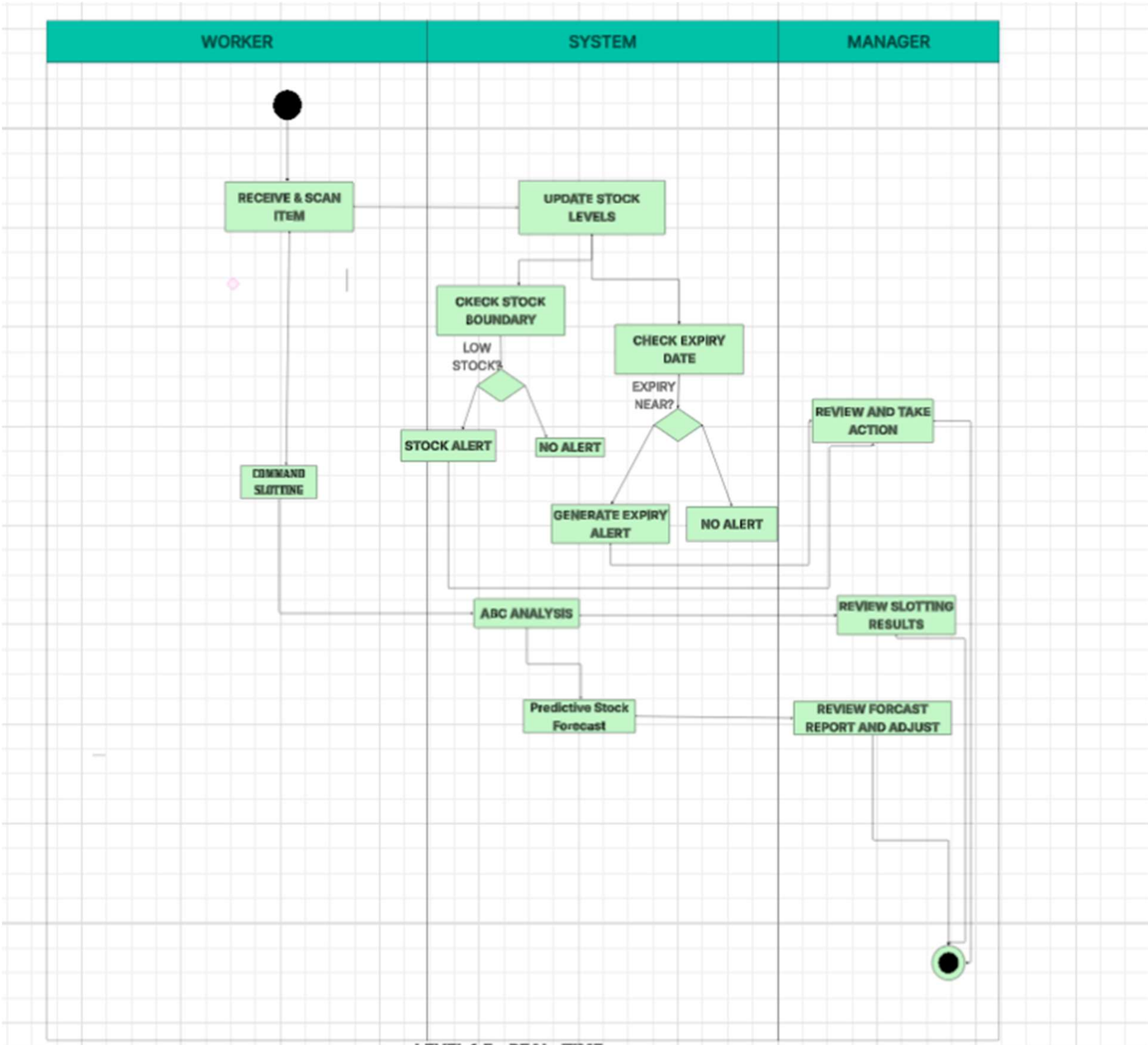
Level:1.1 -
User
Authentication and
Authorization



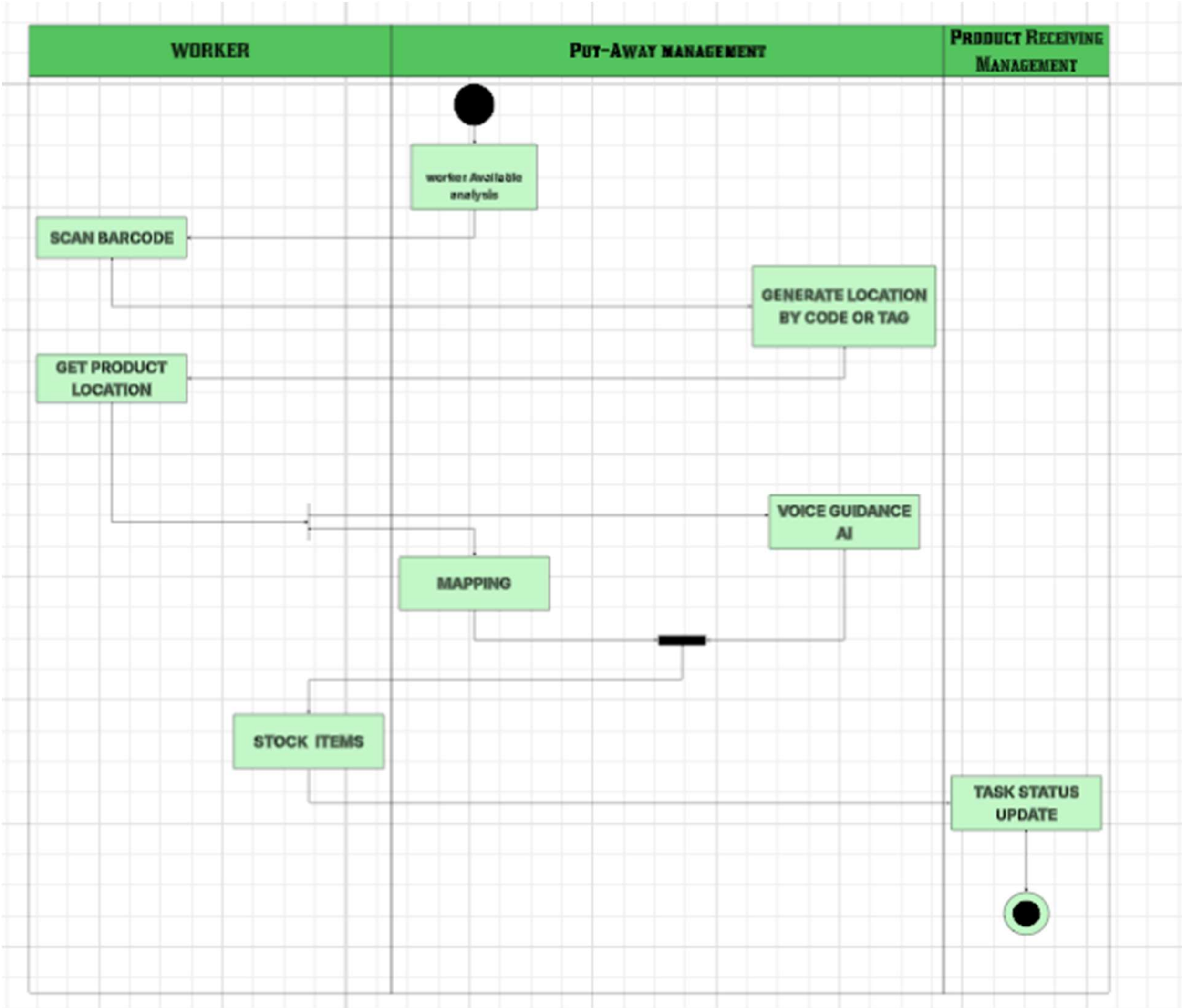
LEVEL:1.2- ORDER PLACEMENT
SWIMLINE



Level:1.3- Item Receiving Management



LEVEL 1.5- REAL -TIME INVENTORY TRACKING



1.6- Worker Mobile App

Data-based Modeling

Noun Table:

Serial	Noun	Problem(p) Solution(s) space	Attribute
1	WMS	p	
2	Warehouse	p	3,4,12,18
3	Warehouse Manager	p	8, 26,28
4	Worker	p	8,9,28,32,33
5	System Administrator / Admin	p	22,23,34,25,27,33
6	Customer	p	10,14,19
7	Supplier	p	18,9,2
8	Inventory/ Stock	p	9
9	Item	p	2,15,`7,19
10	Receiving	p	
11	Receiving Report	s	
12	Storage Location	p	9,8,20
13	Barcode	s	9
14	Order	p	2,6,9,35
15	Pick List	s	
16	Order Picking	p	
17	Packing	p	
18	Shipment	p	14,17,21
19	Returns	p	
20	Inventory	p	
21	Shipping Label	s	
22	FIFO/LIFO	s	
23	Dashboard	s	

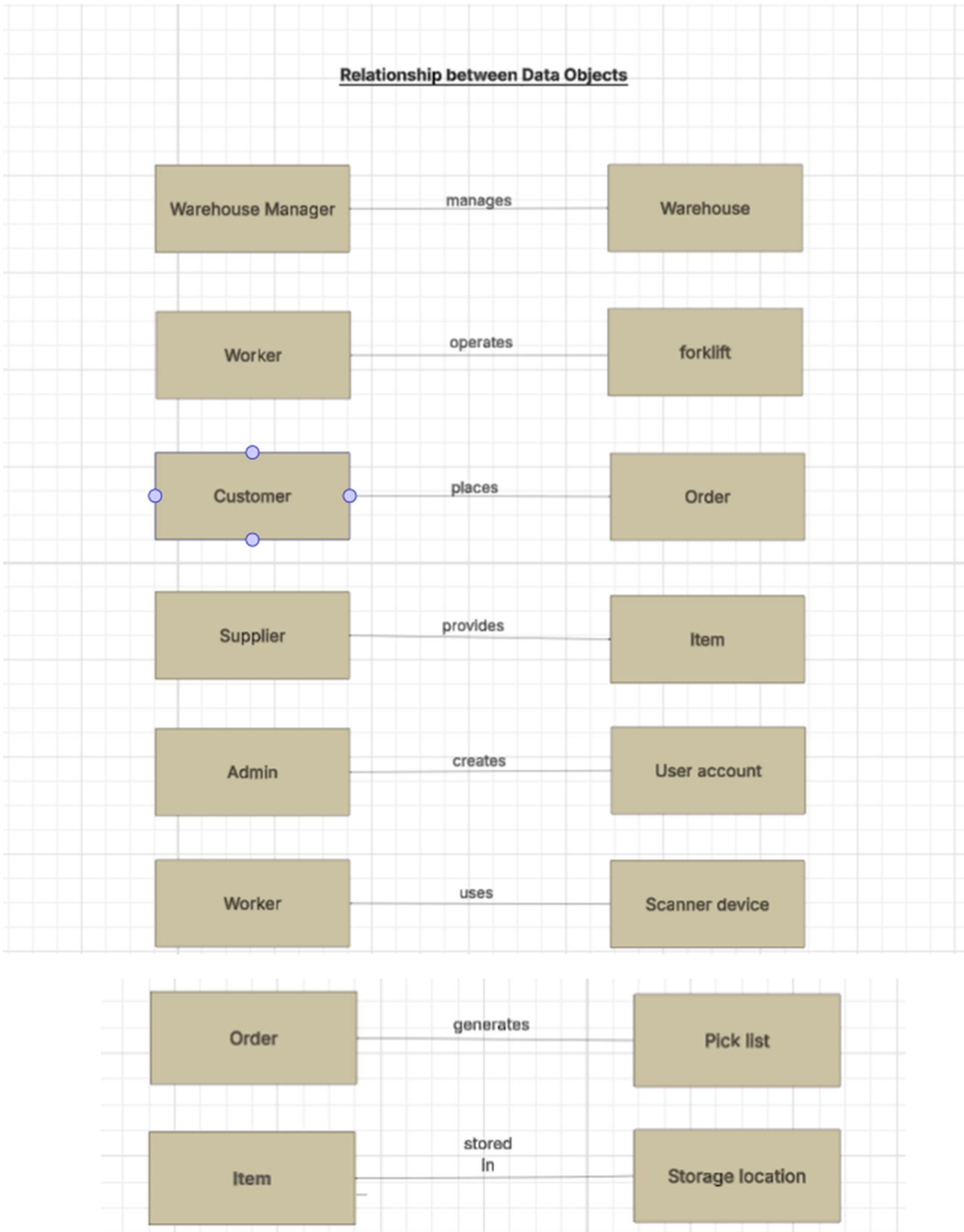
24	Analytics	s	
25	Report	s	
26	Accounting System	s	24,25
27	Database	s	
28	User Account	s	5,6
29	Access Control	s	
30	Warehouse Layout	p	
31	Scanner Device	s	9,13,21
32	Training	p	
33	Mobile Application	s	35,37,38,39,40,41
34	Delivery man	s	14,16,42
35	Menu	s	
36	Item no.	s	
37	Payment	s	
38	login	s	
39	Process	p	10,16,17,,52
40	App Id	s	
41	Version	s	
42	Memo	s	
43	Stock Adjustment	p	8,20,42
44	Storage Location	p	45,46,79
45	Slotting	s	
46	Location Mapping	s	
47	Put-Away	p	9,12,31
48	Order Status	s	
49	Pick List	s	8,9,35
50	Voice Command	s	

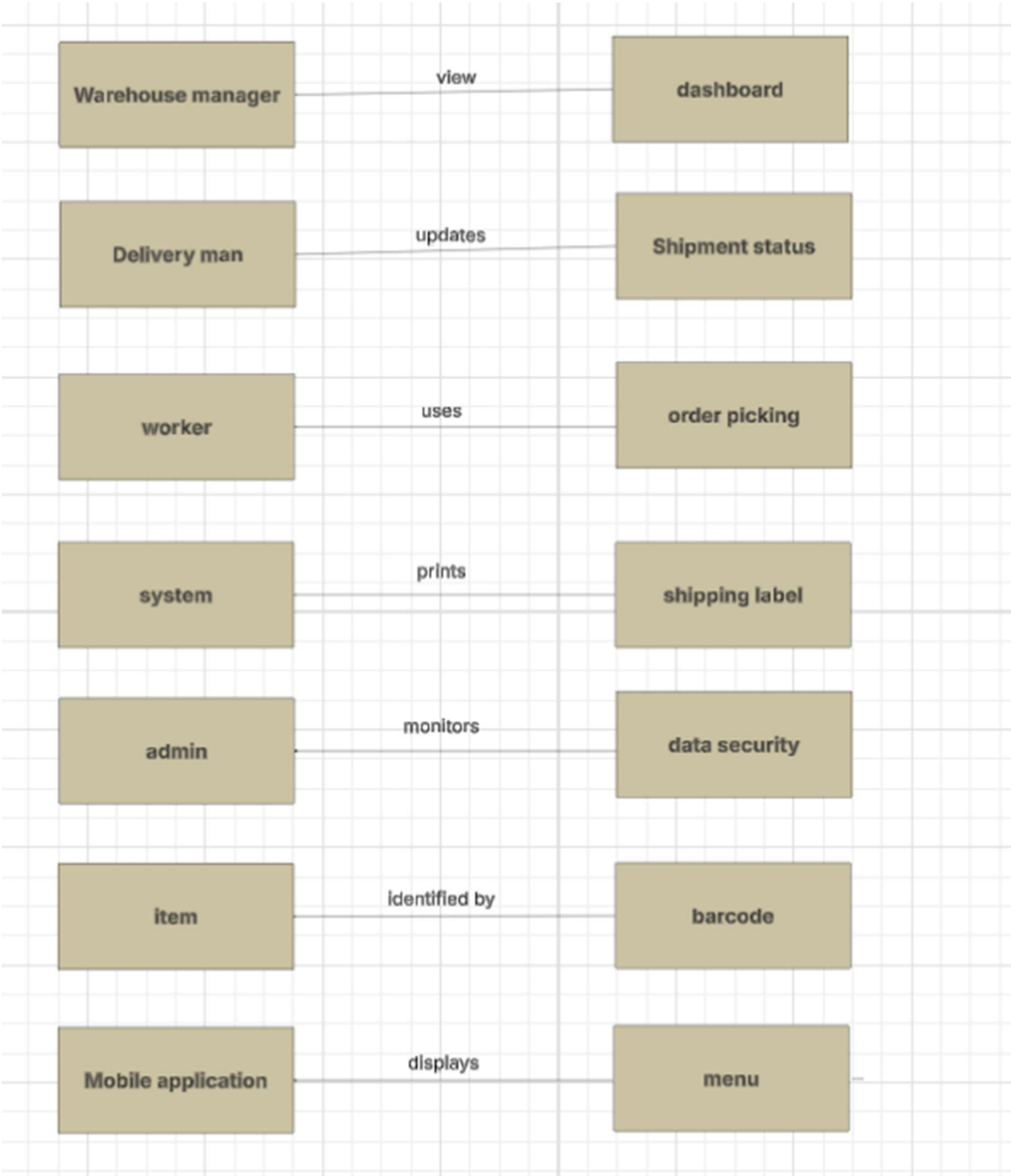
51	Picking Confirmation	s	
52	Dispatch	p	18,34,53
53	Shipment Status	s	2,16,21
54	Returns	p	6,19,85
55	Analytics	s	
56	Inventory Analytics	s	
57	Shortage Alert	s	
58	Operational Report	s	
59	Financial Report	s	
60	Accounting System	s	
61	Invoicing	s	
62	API	s	
63	Scanner Device	s	9,13,21
64	Training	p	
65	User Manual	s	
66	Traceability	s	
67	Sustainability	s	
68	Route Optimization	s	
69	Forklift	p	4,12,32
70	Compliance	s	
71	Data Security	s	2,3,66,67
72	System Update	s	
73	Feasibility Study	s	
74	Data Migration	s	
75	Technical Feasibility	s	
76	Economic Feasibility	s	
77	Operational Feasibility	s	

78	Legal Feasibility	s	
79	Cold Storage	p	9,12,44
80	Hazardous Storage	p	9,12,44
81	Receiving Schedule	s	
82	Appointment	p	7,10,81
83	Delivery Vehicle	p	34,52,84
84	Driver	p	34,83
85	Damage Report	p	9,19,43
86	Exception Handling	s	
87	Stock Reservation	s	
88	Allocation	s	
89	Packing Material	p	17,90
90	Weight	p	9,17,18

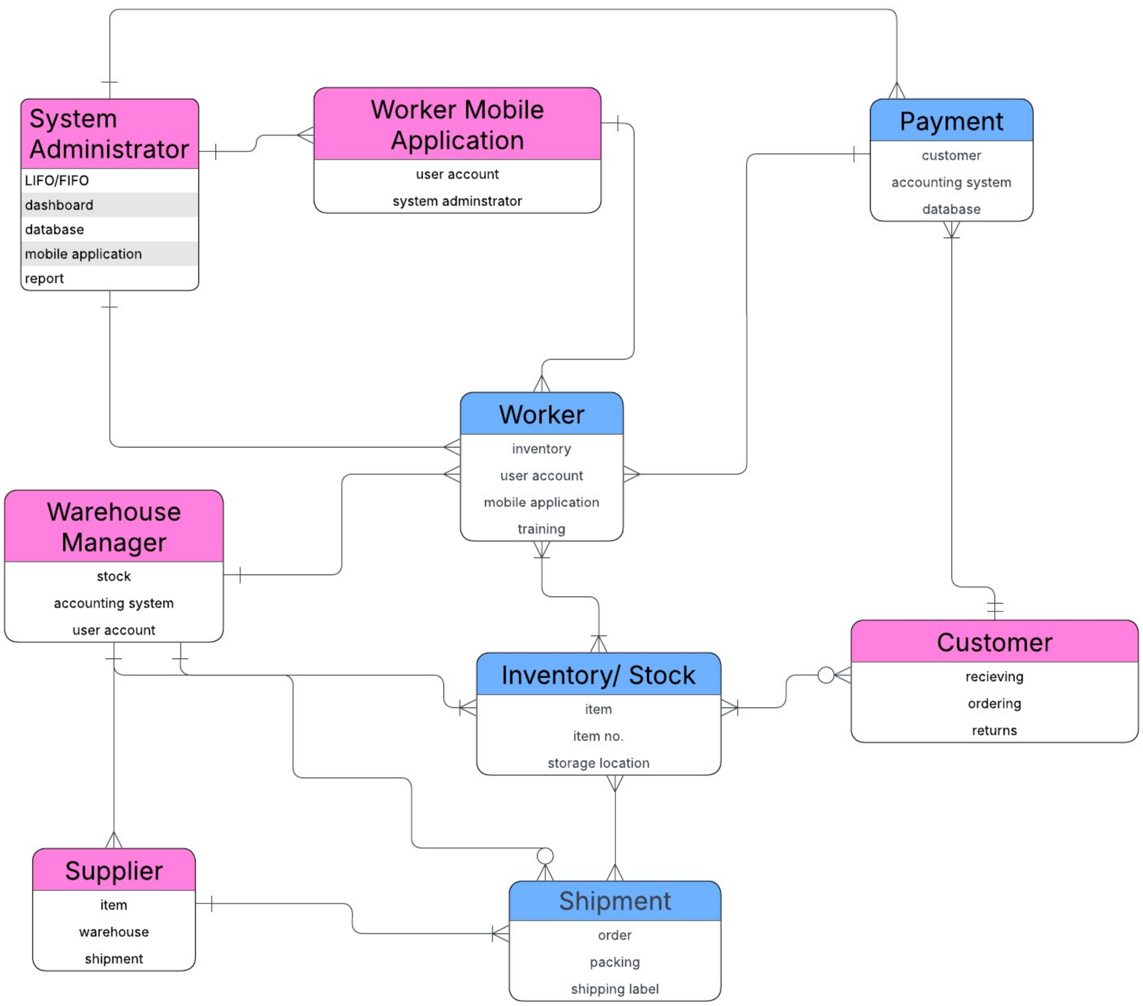
Final data objects :

1. Warehouse
2. Warehouse manager
3. Worker
4. Admin
5. Supplier
6. Item
7. Storage location
8. Barcode
9. Order
10. Shipment
11. User account
12. Scanner device
13. Mobile application
14. Delivery man
15. Menu
16. Pick list
17. Forklift
18. Data security

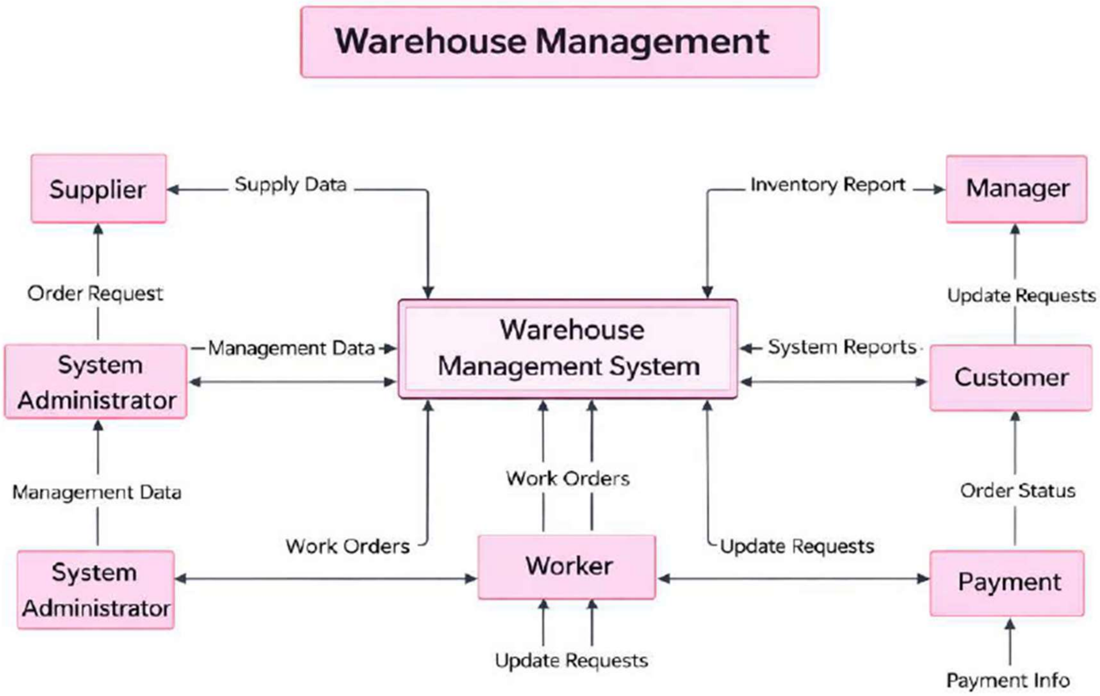

Relationship between Data Objects
Warehouse Manager
manages
Warehouse
Worker
operates
forklift
Customer
places
Order
Supplier
provides
Item
Admin
creates
User account
Worker
uses
Scanner device
Order
generates
Pick list
Item
stored in
Storage location


Warehouse manager
view
dashboard
Delivery man
updates
Shipment status
worker
uses
order picking
system
prints
shipping label
admin
monitors
data security
item
identified by
barcode
Mobile application
displays
menu

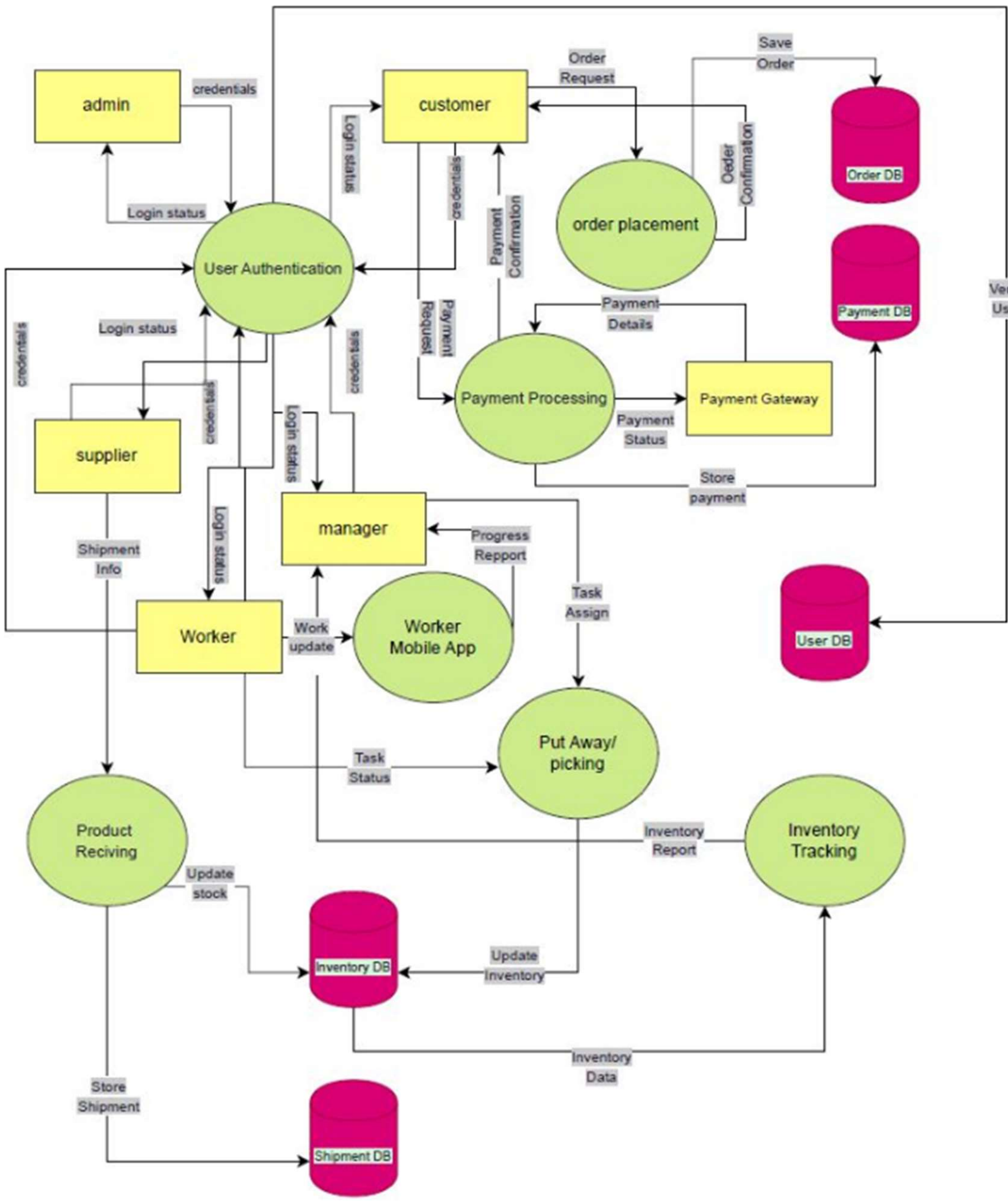
ER- diagram


System Administrator
LIFO/FIFO
dashboard
database
mobile application
report
Worker Mobile Application
user account
system adminstrator
Payment
customer
accounting system
database
Worker
inventory
user account
mobile application
training
Warehouse Manager
stock
accounting system
user account
Inventory/ Stock
item
item no.
storage location
Customer
recieving
ordering
returns
Supplier
item
warehouse
shipment
Shipment
order
packing
shipping label

Data-Flow:



Level-0 :WMS



<u>Level-1 :WMS</u>

Class-based Modeling

Verb Table:

Serial no.	Verb
1.	Login
2.	Logout
3.	Create
4.	Manage
5.	Assign
6.	Authenticate
7.	Receive
8.	Scan
9.	Verify
10.	Inspect
11.	Approve
12.	Reject
13.	Generate
14.	Track
15.	Update
16.	Assign
17.	Store
18.	Adjust
19.	Count
20.	Pick
21.	Confirm
22.	Guide
23.	Pack
24.	Label
25.	Dispatch

26.	Update
27.	Track
28.	Receive
29.	Restock
30.	Dispose
31.	Analyze
32.	Forecast
33.	Monitor
34.	Integrate
35.	Synchronize
36.	Reconcile

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9	Item	p	2,15,`7,19
10	Receiving	p	
11	Receiving Report	s	
12	Storage Location	p	9,8,20
13	Barcode	s	9
14	Order	p	2,6,9,35
15	Pick List	s	
16	Order Picking	p	
17	Packing	p	
18	Shipment	p	14,17,21
19	Returns	p	
20	Inventory	p	
21	Shipping Label	s	
22	FIFO/LIFO	s	
23	Dashboard	s	

24	Analytics	s	
25	Report	s	
26	Accounting System	s	24,25
27	Database	s	
28	User Account	s	5,6
29	Access Control	s	
30	Warehouse Layout	p	
31	Scanner Device	s	9,13,21
32	Training	p	
33	Mobile Application	s	35,37,38,39,40,41
34	Delivery man	s	14,16,42
35	Menu	s	
36	Item no.	s	
37	Payment	s	
38	login	s	
39	Process	p	10,16,17,,52
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47	Put-Away	p	9,12,31
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54	Returns	p	6,19,85
55	Analytics	s	
56	Inventory Analytics	s	
57	Shortage Alert	s	
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59	Financial Report	s	
60	Accounting System	s	
61	Invoicing	s	
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72	System Update	s	
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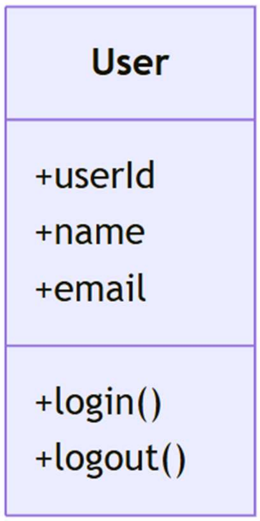
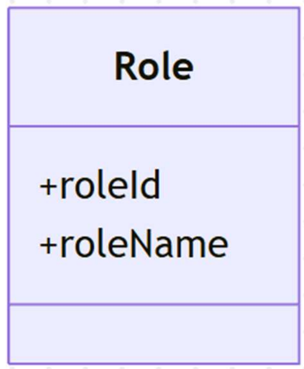
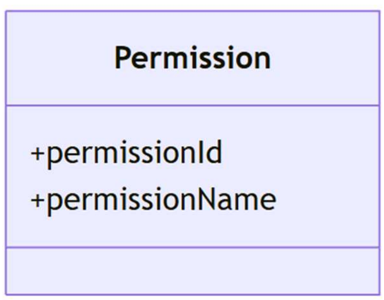
74	Data Migration	s	
75	Technical Feasibility	s	
76	Economic Feasibility	s	
77	Operational Feasibility	s	

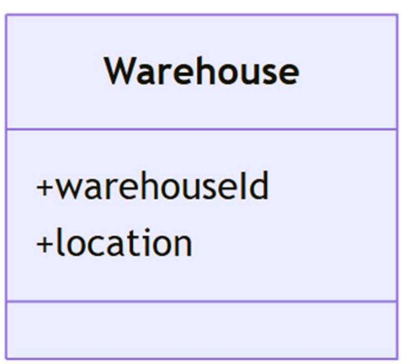
78	Legal Feasibility	s	
79	Cold Storage	p	9,12,44
80	Hazardous Storage	p	9,12,44
81	Receiving Schedule	s	
82	Appointment	p	7,10,81
83	Delivery Vehicle	p	34,52,84
84	Driver	p	34,83
85	Damage Report	p	9,19,43
86	Exception Handling	s	
87	Stock Reservation	s	
88	Allocation	s	
89	Packing Material	p	17,90
90	Weight	p	9,17,18

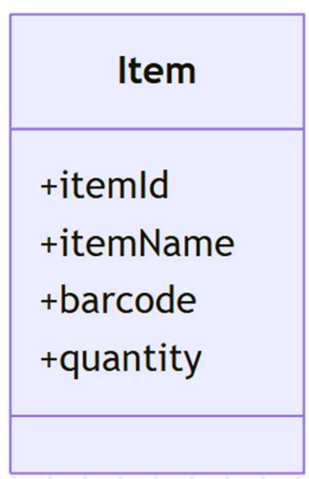
Attribute and Method Identification:

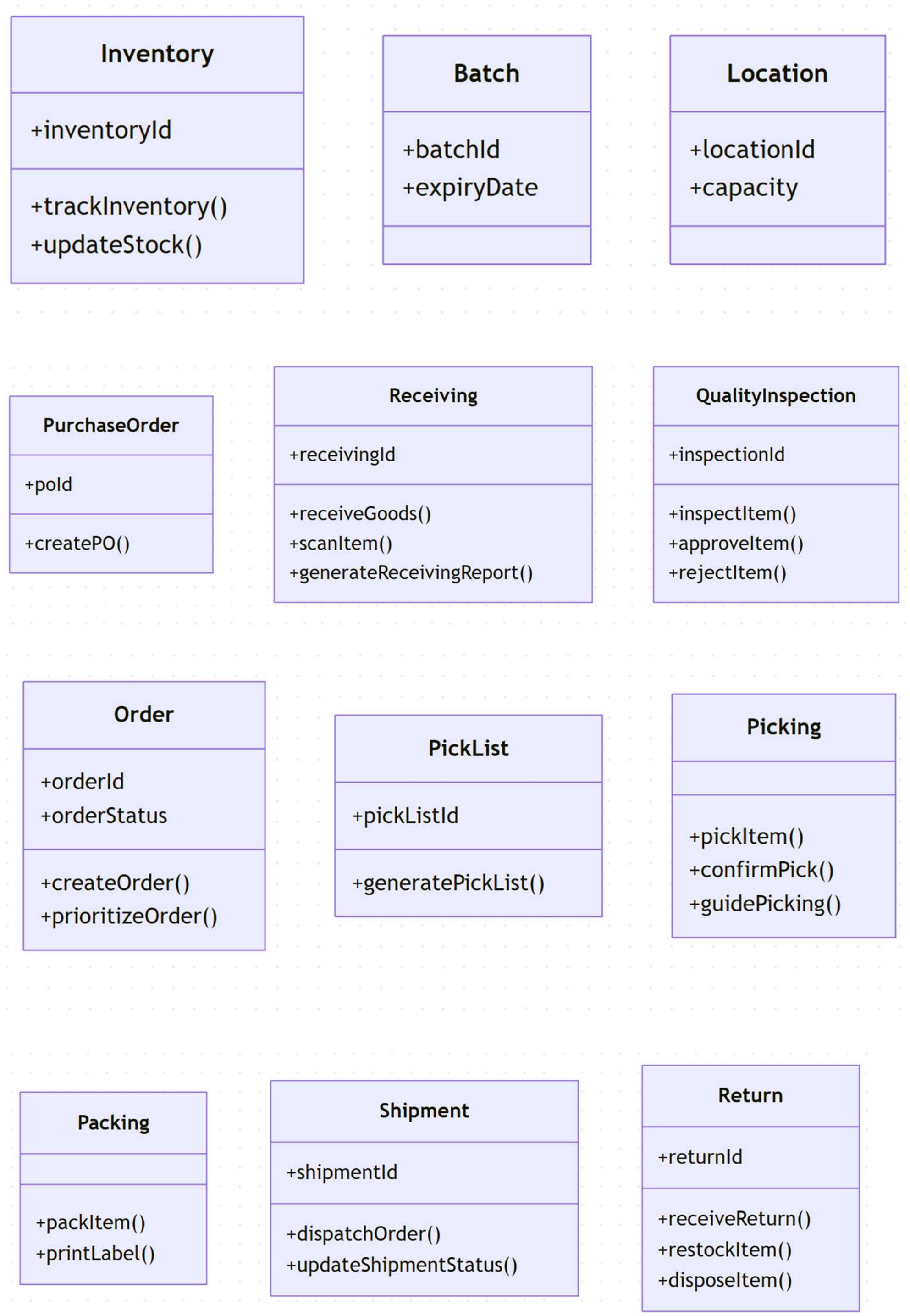


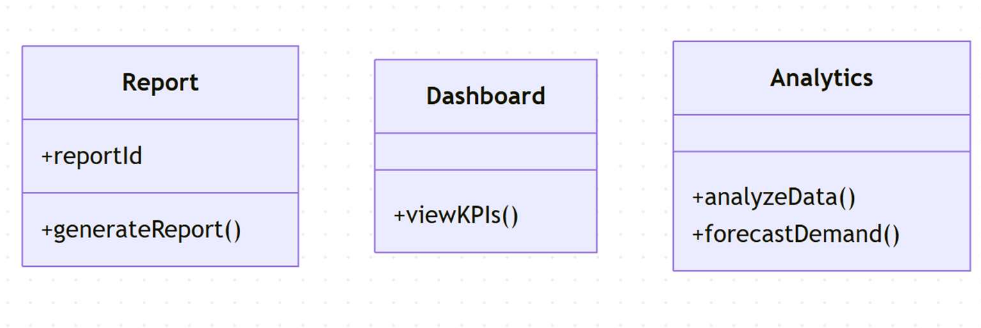


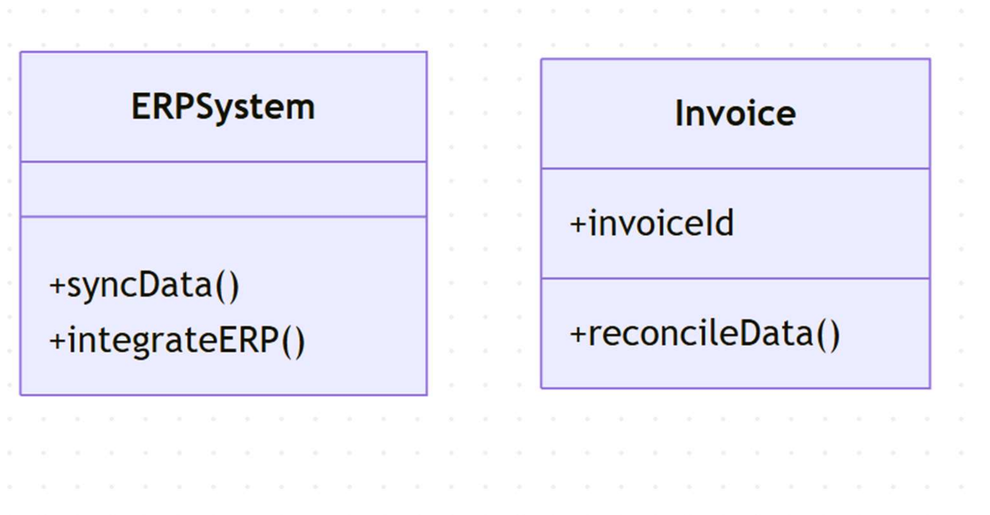







Inventory
+inventoryId
+trackInventory()
+updateStock()
Batch
+batchId
+expiryDate
Location
+locationId
+capacity
PurchaseOrder
+poId
+createPO()
Receiving
+receivingId
+receiveGoods()
+scanItem()
+generateReceivingReport()
QualityInspection
+inspectionId
+inspectItem()
+approveItem()
+rejectItem()
Order
+orderId
+orderStatus
+createOrder()
+prioritizeOrder()
PickList
+pickListId
+generatePickList()
Picking
+pickItem()
+confirmPick()
+guidePicking()
Packing
+packItem()
+printLabel()
Shipment
+shipmentId
+dispatchOrder()
+updateShipmentStatus()
Return
+returnId
+receiveReturn()
+restockItem()
+disposeItem()


Report
+reportId
+generateReport()
Dashboard
+viewKPIs()
Analytics
+analyzeData()
+forecastDemand()


ERPSystem
+syncData()
+integrateERP()
Invoice
+invoiceId
+reconcileData()

CRC Diagram

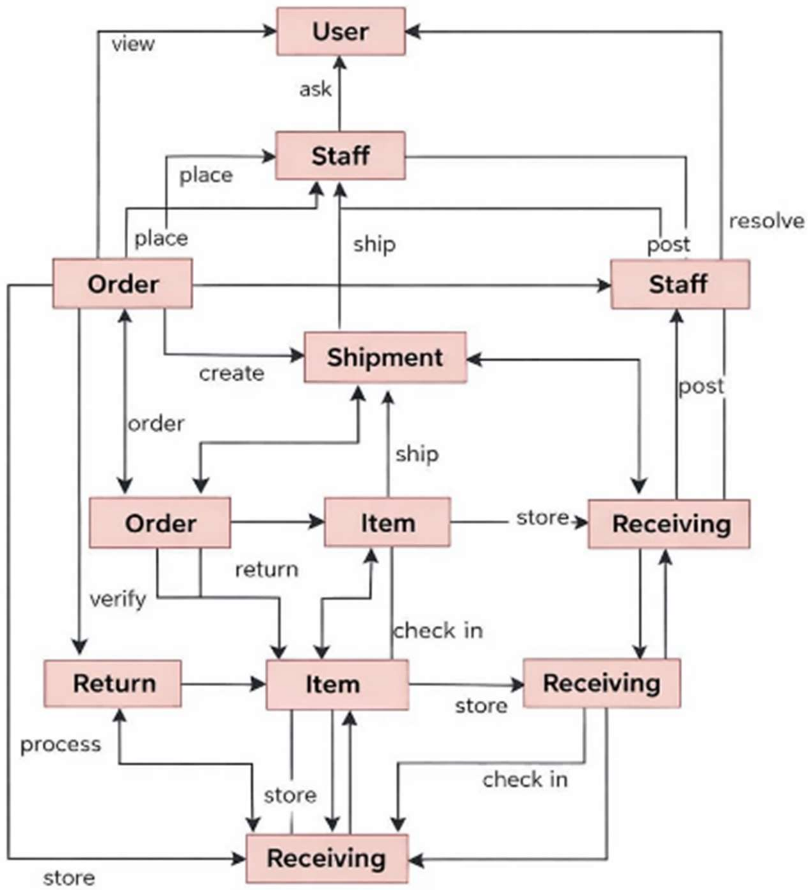


Figure: CRC Diagram for Warehouse Management Classes

Sequence Diagram